ESE 4025 Complete Study Pack

HVAC&R System Design · KUET · past papers 2020, 2021, 2023, 2024 · answers from the course slides

24-hour priority list

Work top to bottom. Blocks 1 to 4 take about 14 hours, which leaves time for sleep, meals and a final read of the one-shot notes. **Chance** is how often the topic came up, with newer papers counting more (2024 ×4, 2023 ×3, 2021 ×2, 2020 ×1): 80% or more is **Very likely**, 60-79% **Likely**, below that **Possible**. **Level** is how hard it is to learn from zero. Inside each section, the pack below follows this same order.

Exception worth knowing: A4 (the R22 80 kW numerical) shows as only Likely, but the 2021 and 2024 papers asked it word for word for 13-15 marks. If Block 3 goes well, do A4 next. You answer 3 of 4 questions in each section, so Blocks 1-3 already cover enough of both sections to pass.

Block 1: very likely and easy. Quick marks first · about 4.2 h (hours 0.0 to 4.2)

#	Topic	Section	Chance	Marks so far	Level	Time
1	**B1** Psychrometric definitions	Section B	Very likely · 100%	52	Easy	25 min
2	**B4** Thermal comfort and the ASHRAE comfort chart	Section B	Very likely · 100%	49	Easy	25 min
3	**A3** Ton of refrigeration, COP and ice / cold-storage load	Section A	Very likely · 100%	43	Easy	25 min
4	**B5** Indoor air quality (IAQ) and ventilation	Section B	Very likely · 100%	43	Easy	25 min
5	**A7** Compressors: hermetic sealed, classification	Section A	Very likely · 100%	37	Easy	25 min
6	**A6** Refrigerant selection: environmental and safety properties	Section A	Very likely · 100%	33	Easy	25 min
7	**A13** Thermoelectric, magnetocaloric and other novel cooling	Section A	Very likely · 90%	35	Easy	25 min
8	**A2** Effect of evaporator and condenser temperature on COP	Section A	Very likely · 90%	34	Easy	25 min
9	**B2** Psychrometric processes and the air washer	Section B	Very likely · 90%	22	Easy	25 min

10	**A1** Vapour compression refrigeration (VCR) cycle	Section A	Very likely · 80%	34	Easy	25 min

Block 2: very likely, medium effort · about 3.8 h (hours 4.2 to 7.9)

#	Topic	Section	Chance	Marks so far	Level	Time
11	**A8** Vapour absorption (VAR): single, double, triple effect	Section A	Very likely · 100%	66	Medium	45 min
12	**B6** Air-conditioning systems: zone, all-air CAV / VAV, dual duct	Section B	Very likely · 100%	63	Medium	45 min
13	**A5** Multistage / multipressure systems and flash cooler	Section A	Very likely · 100%	59	Medium	45 min
14	**B9** Duct sizing numerical (equal friction method)	Section B	Very likely · 100%	56	Medium	45 min
15	**B7** VRF systems (two-pipe / three-pipe heat recovery)	Section B	Very likely · 100%	47	Medium	45 min

Block 3: very likely numericals (hard but they repeat) · about 2.0 h (hours 7.9 to 9.9)

#	Topic	Section	Chance	Marks so far	Level	Time
16	**B8** Cooling load: components, assumptions, numerical	Section B	Very likely · 100%	54	Hard	60 min
17	**B3** Summer AC with ventilation air and bypass factor (numerical)	Section B	Very likely · 90%	41	Hard	60 min

Block 4: likely. Backup for your third answer · about 3.8 h (hours 9.9 to 13.7)

#	Topic	Section	Chance	Marks so far	Level	Time
18	**A10** Performance of a practical plant: EER, SEER, kW/ton	Section A	Likely · 70%	31	Easy	25 min
19	**A15** Cooling towers and condensers	Section A	Likely · 70%	20	Easy	25 min
20	**B10** Duct design rules, outlets, dampers, diffusers	Section B	Likely · 60%	35	Easy	25 min

21	**A12** Air refrigeration and aircraft cooling	Section A	Likely · 60%	33	Medium	45 min
22	**A9** LiBr-water practical problems (crystallisation)	Section A	Likely · 60%	15	Medium	45 min
23	**A4** R22 VCR cycle numerical (with table / p-h chart)	Section A	Likely · 70%	40	Hard	60 min

Only if time is left: asked rarely · about 2.9 h

#	Topic	Section	Chance	Marks so far	Level	Time
24	**A14** Thermostatic expansion valve	Section A	Possible · 50%	22	Easy	25 min
25	**B11** Sol-air temperature, window solar gain, infiltration	Section B	Possible · 50%	19	Medium	45 min
26	**A16** Linde air liquefaction	Section A	Possible · 20%	10	Medium	45 min
27	**A11** Ammonia ice-plant numerical (COP % of theoretical)	Section A	Possible · 50%	27	Hard	60 min

CONTENTS

1. One-shot notes
2. **Section A: Refrigeration**: A3 Ton of refrigeration, COP and ice / cold-storage load, A7 Compressors: hermetic sealed, classification, A6 Refrigerant selection: environmental and safety properties, A13 Thermoelectric, magnetocaloric and other novel cooling, A2 Effect of evaporator and condenser temperature on COP, A1 Vapour compression refrigeration (VCR) cycle, A8 Vapour absorption (VAR): single, double, triple effect, A5 Multistage / multipressure systems and flash cooler, A10 Performance of a practical plant: EER, SEER, kW/ton, A15 Cooling towers and condensers, A12 Air refrigeration and aircraft cooling, A9 LiBr-water practical problems (crystallisation), A4 R22 VCR cycle numerical (with table / p-h chart), A14 Thermostatic expansion valve, A16 Linde air liquefaction, A11 Ammonia ice-plant numerical (COP % of theoretical)
3. **Section B: Air conditioning**: B1 Psychrometric definitions, B4 Thermal comfort and the ASHRAE comfort chart, B5 Indoor air quality (IAQ) and ventilation, B2 Psychrometric processes and the air washer, B6 Air-conditioning systems: zone, all-air CAV / VAV, dual duct, B9 Duct sizing numerical (equal friction method), B7 VRF systems (two-pipe / three-pipe heat recovery), B8 Cooling load: components, assumptions, numerical, B3 Summer AC with ventilation air and bypass factor (numerical), B10 Duct design rules, outlets, dampers, diffusers, B11 Sol-air temperature, window solar gain, infiltration

★ One-shot notes

Section A: Refrigeration

Units and basics

Refrigeration: removing heat from a substance under controlled conditions and keeping it below the surroundings' temperature.

1 TR = melting 1 tonne ice from and at 0 °C in 24 h = 232.6 kJ/min; practice: **210 kJ/min = 3.5 kW**.

```
COP = RE/W = (h1 - h4)/(h2 - h1); COP_HP
= COP_R + 1
Carnot COP_R = TL/(TH - TL)
```

Refrigerator: wants Q_L (cold side). Heat pump: wants Q_H (hot side). Refrigeration controls temperature only; AC controls temperature + humidity + air motion + purity.

Ice / product load recipe

```
Q = m[cΔT + latent] / time(min) ÷ 210 =
TR
ice latent 335 kJ/kg, water c = 4.187, 1
t ice/day = 3.88 kW
```

2000 kg, 0→45 °C: 3.46 TR · 1500 kg, 0→25 °C: 2.18 TR · 600 kg fruit 25→−5 °C in 8 h (c 1.255, L 105): 0.85 TR

VCR cycle (C-C-E-E)

1→2	Compressor, isentropic, w = h2 − h1
2→3	Condenser, const p, q = h2 − h3
3→4	Expansion valve, h4 = h3 (throttling)
4→1	Evaporator, RE = h1 − h4

Throttle replaces turbine; dry vapour compressed (Carnot impractical: wet compression, wet expansion).

Dry sat. after compression: s1 = s2 = s_{g2}; x1 = (s1 − s_{f1})/(s_{g1} − s_{f1}); h1 = h_{f1} + x1h_{fg1}.

Subcooling: RE ↑, W same → COP ↑ always.
Superheat: RE ↑ and W ↑; COP may go either way; protects compressor.

Effect of temperatures

T_{evap} ↑ → RE ↑, W ↓, v1 ↓ (capacity ↑) → COP ↑.
T_{cond} ↑ → RE ↓, W ↑, discharge temp ↑ → COP ↓.

Evaporator temp has more influence: ∂COP/∂Te = Tc/ΔT² > Te/ΔT². R22 example: +5 K evap → 4.13, −5 K cond → 4.11 (base 3.61).

R22 numerical recipe

```
find p_e, p_c; h1 (with superheat), v1,
s1
h2 at p_c with s2 = s1; h3 = h_f(Tc -
subcool) = h4
m = Q/(h1-h4); V = m·v1; P = m(h2-h1)
```

80 kW, −10/45 °C, 5 K sub, 10 K sup: h1 408.2, h2 450.4, h4 249.6, v1 0.0686 → m 0.505 kg/s, **V 0.0346 m³/s, COP 3.75, P 21.3 kW**.

2020 table (50 kW, 35/−10, h2 435.2): h1 401.06, h3 243.14 → RE 157.9, m 0.317, P 10.8 kW, COP 4.63.

Ammonia ice plant recipe

Q = tonnes × 3.88 kW. x1 from s, h1, COP_{th} = (h1 − h_{f3})/(h2 − h1), × %, P = Q/COP.

25 t, 15/−5, 45%: COP_{th} 12.8 → **16.9 kW**. 30 t, 20/−10, 55%: 8.15 → **26.0 kW**. Slide: 30 t, 25/−15, 60% → 33.2 kW.

Multistage

When: T_c − T_e > 40 K (Gosney), high pressure ratio. Single stage then has: big cylinder, high discharge temp (oil burns), low η_{vol}, high friction/cost.

Advantages: less work, better η_{vol}, low discharge temp, more RE, less leakage, uniform torque, cheaper compressor. **Disadvantages:** cost, complexity.

```
p_i = √(p_c · p_e)
```

Flash cooler: LP comp 1→2, mix with flash gas → 3, HP comp → 4, condense → 5, valve 1 → 6 in flash chamber, vapour 7 to HP, liquid 8 → valve 2 → 9 evaporator. Flash gas skips evaporator and LP stage.

Single comp + multiple exp. valves: E3→E2→E1 in series, back-pressure valves, vapours mix to one compressor. **Individual compressors:** each evaporator has own compressor; COP higher.

Refrigerants

Criteria: thermodynamic, environmental & safety, economic.

Thermo: suction p above atm, low discharge p, low pressure ratio, high latent heat, low isentropic index, low liquid c_p, high vapour c_p, high k, low viscosity.

Environment: ODP = 0 (R11 = 1, Cl/Br destroy ozone), low GWP (CO_2 = 1), low TEWI (direct leakage + indirect energy CO_2).

Safety: non-toxic, non-flammable, chemically stable, compatible, oil miscible (NH_3 no, R12 yes, R22 partly), easy leak detection. ASHRAE A/B toxicity, 1/2/3 flammability.

Compressors

Classes: method (recip, rotary, centrifugal), strokes (single/double acting), stages, drive (direct/belt), prime mover (open, semi-hermetic, hermetic).

Hermetic: motor + compressor on one shaft in one welded shell; no shaft seal. +: no leakage, quiet, compact, simple lubrication. −: inaccessible, separate pump for charging, burnout contaminates.

Centrifugal vs recip +: no valves/pistons (long life), no vibration, quiet, 3000+ rpm direct drive, big volumes, 50-5000 TR, high efficiency, less floor area. −: surging <35% load, not below 50 TR.

Absorption (VAR)

Compressor replaced by absorber + pump + generator (heat input). Pairs: H_2O-LiBr (AC, >0 °C), NH_3-H_2O (below 0 °C).

	Single	Double	Triple
Generators	1	2 (HTG vapour heats LTG)	3
COP	0.7	1.0-1.2	1.5-1.7
Heat	80-120 °C	150-170 °C	200+ °C

VAR vs VCR: heat vs work; low vs high COP; few vs many moving parts; part load fine vs drops; quiet; bulky small units; good where waste heat is cheap.

Heat sources: steam, hot water, direct gas/oil firing, waste heat/exhaust, solar, geothermal.

LiBr problems: crystallization, air leakage (vacuum, purge), corrosion (inhibitors), freezing below ~4 °C, large vapour volume.

Practical plant performance

Full load: COP, EER, kW/ton. Part load: SEER, IPLV/NPLV.

```
EER = Btu/h ÷ W; COP = EER/3.412; kW/ton = 12/EER
IPLV = 0.01A + 0.42B + 0.45C + 0.12D (100/75/50/25%)
```

HSPF = heating season Btu ÷ Wh (heat pumps). Energy Star = EPA efficiency label.

Other short answers

Air cycle: compress → cool → expand in turbine → absorb; COP = $1/(r_p^{(\gamma-1)/\gamma} - 1)$. Aircraft (2020): Q = 50 × 1 × 35 = 1750 kJ/min = **8.33 TR**, COP ≈ 0.19.

TEV: bulb pressure opens; evaporator + spring pressure close; constant superheat.

Thermoelectric: Peltier, p-n couples; reverse current swaps hot and cold; COP 0.3-0.7. Magnetocaloric not commercial: costly magnets and Gd, small ΔT, low power density.

Cooling tower: natural/mechanical (forced, induced), counter/cross-flow. Range = in − out; Approach = out − WBT; ε = R/(R + A).

Linde: compress ~200 bar, HX regenerative cooling, J-T throttling, separate liquid; y = (h1 − h2)/(h1 − h_f).

Section B: Air conditioning

Psychrometric definitions

DBT	ordinary thermometer temperature
WBT	lowest temp by evaporating water; wet wick in moving air; DBT − WBT = depression
DPT	temp where vapour starts condensing on cooling at constant p (constant w)
w	kg vapour/kg dry air = 0.622 $p_v/(p_t - p_v)$
φ	p_v/p_{sat} at same DBT
μ	w/w_s at same T, p (0 dry, 1 saturated)
c_{pm}	1.005 + 1.88w ≈ 1.0216 kJ/kg K (heat to raise 1 kg d.a. + vapour by 1 K)
h	1.005t + w(2501 + 1.88t)
SHF	$Q_s/(Q_s + Q_L)$; 0.75-0.8 dry climate, 0.6 humid
BPF X	$(t_c - t_s)/(t_o - t_s)$; CF = 1 − X; lower with more rows, lower velocity, closer fins
ADP	effective coil surface temp; SHF line meets saturation curve

Saturated air: DBT = WBT = DPT. Always DBT ≥ WBT ≥ DPT.

SHF line: join chart reference point (26 °C, 50%) to SHF on protractor, draw parallel through room point. tanθ = (1/2451)(1 − SHF)/SHF.

Processes and air washer

Sensible heat/cool: w constant (coil above DPT). Cool + dehumidify: coil below DPT; $m_w = m_a(w_o - w_c)$, $Q = m_a(h_o - h_c)$. Heat + humidify: heating coil then steam/spray.

```
Qs = m·cpm·Δt   QL = m·2501·Δw   Qt = m·Δh
```

t_w	Air washer process
< DPT	cool + dehumidify (cool water)
DPT-WBT	cool + humidify (cool water)
= WBT	adiabatic/evaporative, h const (recirculate)
WBT-DBT	cool + humidify, h ↑ (heat water)
> DBT	heat + humidify (heat water)

Air washer = spray chamber + eliminators + pump + cooler/heater + make-up water. $\eta_H = 1 - X$.

Summer AC + ventilation + BPF recipe

```
mix: t1 = r·ti + (1-r)to (same for w, h)
coil: ts = ADP + X(t1 - ADP)
m_d = RSH / [1.02(ti - ts)]   m_o = (OA share)·m_d
OASH = m_o·1.02(to-ti)  OALH = m_o·2500(wo-wi)
GTH = RSH + RLH + m_o(ho - hi) ; TR = GTH/3.5
```

RSHF line: room i → supply. GSHF line: mixed 1 → coil ADP. ADP unknown? Try ADP until 1.02(ti − ts)/[2501(wi − ws)] = RSH/RLH.

2024: m_d 1.83 kg/s, OASH 8.4, OALH 8.0, **GTH 41.8 kW**. 2021 (OA 35 °C/65%): ADP 11.5 °C, 46.4 cmm, **91.1 kW**. Slide ex. (4:1, X 0.1): 1.73 kg/s, 37.7 kW.

Thermal comfort

Def: condition of mind expressing satisfaction with the thermal environment. **Factors:** air temp, MRT, RH, air velocity, activity (met), clothing (clo).

Comfort chart: ET lines + 60% RH top + 2 °C DPT bottom. Winter ET 20-23.5 °C, summer 22.5-26 °C, 1.2 met.

ET: temp of standard 50% RH environment giving the same skin heat loss. **Operative:** $(t_a + t_{mrt})/2$, radiation + convection.

```
PPD = 100 - 95 exp(-0.03353PMV&sup4; -
0.2179PMV²); PMV 0 → PPD 5%
```

So nobody can satisfy everyone. Indoor design: 24-26 °C, 50-60%. Outdoor design: statistical 1-2.5% weather values.

IAQ

SBS (symptoms vanish outside) vs BRI (real disease, persists). 90% time indoors; indoor air 2-5× worse.

Control: 1. Source control (low-VOC, intakes 8-10 ft high, 10-15 ft from exhausts, RH < 60%) 2. Filtration (MERV: panel 1-4, pleated 5-8, bag 9-15; HEPA 99.97% at 0.3 µm; ULPA 99.999% at 0.12 µm) 3. Dilution (ventilation, CO_2 sensor).

```
V = n·Np / (C_in - C_out); Np = 0.011 cfm
CO2; C_in 1000 ppm
1 person, 300 ppm out: 15.7 cfm
```

30 students: Khulna (400) **550 cfm = 0.26 m³/s**; Dhaka (800) **1650 cfm = 0.78 m³/s** (3×).

AC systems

Zone: space(s) controlled by one thermostat. **Split:** indoor evaporator + blower, outdoor compressor + condenser, two pipes.

All-air: single-duct CAV, CAV reheat, multizone, dual duct, VAV. +: central plant, good humidity/filtration, economizer, quiet. −: big ducts, fan energy.

Dual duct CAV: one fan, hot + cold ducts, mixing box per zone; constant air, variable temperature.

CAV vs VAV: fixed air/variable temp vs fixed temp/variable air; VAV saves fan energy.

Rooftop: + packaged, saves floor space, quick; − roof load, weather, service access.

Choice matrix (2024): Σ importance × score: split 346, **rooftop 434**.

VRF

Inverter compressor + EEV per indoor unit, one outdoor unit, many indoor units. +: part-load efficiency, zone control, heat recovery, no ducts, modular, quiet.

3-pipe heat recovery: discharge gas, suction gas, liquid. Heating unit: hot gas in → liquid out. Cooling unit: liquid in → suction gas out. **2-pipe:** 2 pipes to a BC controller that separates gas/liquid.

Multi-split vs VRF: few units, own pipes, same mode, short pipes vs up to 60 units, shared branches, simultaneous heat/cool, long pipes, EEV.

Cooling load

Cooling: unsteady, daytime peak, internal gains added. Heating: steady, pre-dawn peak, no sun or internal gains.

Assumptions: statistical outdoor design data, clear sky, full occupancy, equipment at representative capacity, near-max not max.

```
opaque: U·A·CLTD(ETD)   floor/basement:
U·A·(To-Ti)
glass: A[Ug(To-Ti) + SHGF·SC·CLF]
infil: m = ρ·ACH·Vol/3600; m·cpm·ΔT +
m·2501·ΔW
people: N×S×CLF + N×L ; lights:
W×UF×BF(1.25)×CLF
GTH = RSH + RLH + OATH; ×1.25 safety
```

2024 room: RSH 5035 W, RLH 578 W, ×1.25 = **7.0 kW ≈ 2 TR** (glass U 3.12 assumed, inside 25 °C/50%).

Sol-air temp: $t_{sa} = t_o + \alpha I/h_o - \varepsilon \Delta R/h_o$. **Infiltration:** uncontrolled leakage; crack method = crack length × leakage per m; or ACH method.

Duct design

```
Equal friction: same f (mm H2O/m)
everywhere
D = (0.002268 Q^1.852 / f)^(1/4.973)
De = 1.3(ab)^0.625/(a+b)^0.25 ; 3:1 → b =
0.548D, a = 3b
Fan p = f × longest run + ΣK·pv; pv =
1.2V²/(2×9.81) mm
```

Q m³/s	2	3	5	8	10	14
D @0.1	0.605	0.703	0.850	1.013	1.101	1.248
3:1 a×b	.99×.33	1.16×.39	1.40×.47	1.66×.55	1.81×.60	2.05×.68

@0.09 (2020): 8 → 1.035, 6 → 0.930, 3 → 0.718 m. 2021: 12 → 1.94×0.65, 9 → 1.74×0.58, 4 → 1.29×0.43.

Rules: short and straight, low aspect ratio (≤4:1), velocity limits for noise, gradual transitions, sealed and insulated, dampers for balancing.

Outlet location: throw/spread/drop, near heat sources (windows), no short-circuit to returns, noise, ceiling height. **Dampers** control/balance flow; **diffusers** spread and slow supply air.

Exam plan: In each section answer the 3 questions that contain the numericals above (R22 cycle / ice load in A; summer AC with BPF, cooling load and duct sizing in B) plus definition parts. The numericals repeat almost unchanged.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ACH	air changes per hour
ADP	apparatus dew point
AHU	air handling unit
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BPF	bypass factor
CAV	constant air volume
CFC	chlorofluorocarbon
CLF	cooling load factor
CLTD	cooling load temperature difference
COP	coefficient of performance
DBT	dry bulb temperature
DPT	dew point temperature
EER	energy efficiency ratio
ESHF	effective sensible heat factor
ETD	equivalent temperature difference
FCU	fan coil unit
GSHF	grand sensible heat factor
GTH	grand total heat
GWP	global warming potential
HCFC	hydrochlorofluorocarbon
HFC	hydrofluorocarbon
HVAC	heating, ventilation and air-conditioning
IAQ	indoor air quality
IIR	International Institute of Refrigeration
LiBr	lithium bromide
NH3	ammonia
OALH	outside air latent heat
OASH	outside air sensible heat
OATH	outside air total heat
ODP	ozone depletion potential
RE	refrigerating effect
RH	relative humidity
RLH	room latent heat
RSH	room sensible heat
RSHF	room sensible heat factor
RTH	room total heat

SC	shading coefficient
SEER	seasonal energy efficiency ratio
SHGF	solar heat gain factor
TEV	thermostatic expansion valve
TR	ton of refrigeration (3.5 kW)
TSH	total sensible heat
VAR	vapour absorption refrigeration
VAV	variable air volume
VCR	vapour compression refrigeration
VRF	variable refrigerant flow
WBT	wet bulb temperature

Section A: Refrigeration

A3 Ton of refrigeration, COP and ice / cold-storage load Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(a)** [10] Define Ton. Calculate the refrigeration tonnage for 2000 kg ice melting from 0 °C to 45 °C.
- **2023 Q1(d)** [7] Calculate the refrigeration tonnage for 1500 kg ice from 0 °C to 25 °C.
- **2021 Q1(a)** [8] Define Ton and COP (coefficient of performance) of refrigeration system. What is the basic difference between a refrigerator and a heat pump?
- **2020 Q1(a)** [8] What is meant by TR (ton of refrigeration (3.5 kW))? Write down the differences between refrigeration and air conditioning system.
- **2020 Q1(c)** [10] 600 kg of fruits are supplied to a cold storage at 25 °C. The cold storage is maintained at −5 °C and the fruits get cooled to the storage temperature in 8 hours. The latent heat of freezing is 105 kJ/kg and specific heat of fruit is 1.255 kJ/kgK. Find the refrigeration capacity of the plant.

Before you start: ton of refrigeration and ice load

1 TR = cooling needed to freeze (or melt) 1 tonne of ice at 0 °C in 24 hours ≈ 3.5 kW = 210 kJ/min.

Heat to change ice/water: **latent heat** (melting or freezing, no temperature change) = m × 335 kJ/kg; **sensible heat** (temperature change) = m × c × ΔT, with c = 4.187 kJ/kg·K for water, about 2.1 for ice.

Definitions (2020 1a, 2021 1a, 2024 1a)

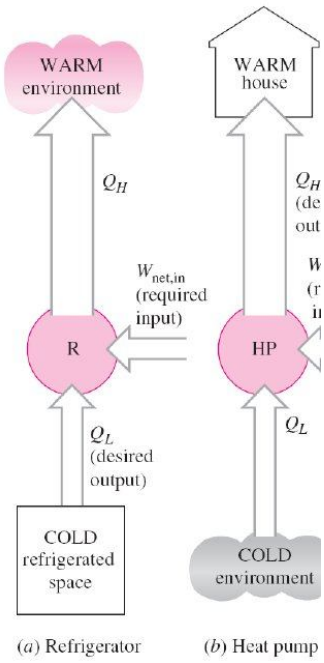
Refrigeration is the process of removing heat from a substance under controlled conditions and keeping it below the temperature of its surroundings.

Ton of refrigeration (TR): the refrigerating effect produced by the uniform melting of 1 tonne (1000 kg) of ice from and at 0 °C in 24 hours.

```
1 TR = 1000 × 335 kJ / 24 h = 232.6 kJ/min ≈ 210 kJ/min = 3.5 kW (value used in practice)
```

COP = heat extracted / work done = refrigerating effect / compressor work = $(h_1 - h_4)/(h_2 - h_1)$

Refrigerator vs heat pump



Refrigerator and heat pump work between the same two temperatures (course slides)

Refrigerator	Heat pump
Purpose: remove heat Q_L from the cold space	Purpose: supply heat Q_H to the warm space
$COP_R = Q_L/W$	$COP_{HP} = Q_H/W = COP_R + 1$
Useful side: evaporator	Useful side: condenser

Same device and same cycle; only the objective differs.

Refrigeration vs air conditioning (2020 1a)

Refrigeration	Air conditioning
Controls only temperature (usually below ambient)	Controls temperature, humidity, air motion and air purity together
Applied to products: food, ice, medicine, process	Applied mainly to human comfort or industrial spaces
Space may be far below 0 °C	Space kept near 20-26 °C
Refrigeration is a part of AC	AC uses refrigeration plus heating, humidifying, filtering, fans

Numerical 1 (2024 1a): 2000 kg ice, 0 °C to 45 °C

> Read as: 2000 kg of ice at 0 °C melts and the water is warmed to 45 °C in 24 h (making ice from water at 45 °C gives the same answer).
> Latent heat = 2000 × 335 = 670 000 kJ
> Sensible heat = 2000 × 4.187 × (45 − 0) = 376 830 kJ
> Total Q = 1 046 830 kJ per day = 1 046 830 / 1440 = 727.0 kJ/min
> **Tonnage = 727.0 / 210 = 3.46 TR**

Numerical 2 (2023 1d): 1500 kg ice, 0 °C to 25 °C

Q = 1500 [335 + 4.187 × 25] = 1500 × 439.7 = 659 550 kJ/day
= 659 550 / 1440 = 458.0 kJ/min
Tonnage = 458.0 / 210 = 2.18 TR

Numerical 3 (2020 1c): 600 kg fruit, 25 °C to −5 °C in 8 h

Sensible heat = m c ΔT = 600 × 1.255 × (25 − (−5)) = 22 590 kJ
Latent heat of freezing = 600 × 105 = 63 000 kJ
Total = 85 590 kJ in 8 h = 85 590 / 480 = 178.3 kJ/min
Capacity = 178.3 / 210 = 0.85 TR

Remember: Load (kJ) ÷ time (min) ÷ 210 = TR. Ice: 335 kJ/kg latent, water 4.187 kJ/kg K.

Same answer, step by step with reasons

Step (write this)	Why / what it means
`2024: Latent = 2000 × 335 = 670,000 kJ`	Melting the ice at 0 °C (no temperature change).
`Sensible = 2000 × 4.187 × 45 = 376,830 kJ`	Warming the water from 0 to 45 °C.
`Total = 1,046,830 kJ per day ÷ 1440 min = 727 kJ/min`	24 h = 1440 min.
`TR = 727 / 210 = 3.46 TR`	1 TR = 210 kJ/min.
`2023: 1500 × (335 + 4.187 × 25) / 1440 / 210 = 2.18 TR`	Same recipe in one line.

Memory trick: "Latent + Sensible, per minute, over 210".

A7 Compressors: hermetic sealed, classification Must study Very likely · 100% Easy

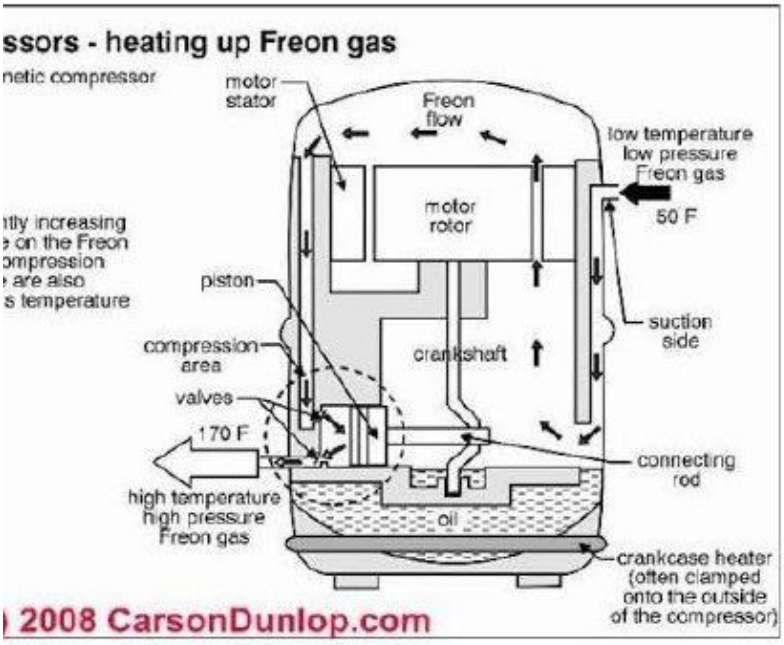
PAST-PAPER QUESTIONS

- **2024 Q3(d)** [5] Write down the advantages and disadvantages of hermetic sealed compressor.
- **2024 Q3(c)** [7] Classify compressors. Write down the advantages of centrifugal compressors over reciprocating compressors.
- **2023 Q3(c)** [8] What is meant by hermetic sealed compressor? Write down its advantages and disadvantages.
- **2021 Q3(d)** [10] What is meant by hermetic sealed compressor? What are the advantages and disadvantages of this compressor?
- **2020 Q3(b)** [7] What is meant by hermetic sealed compressor? What are its advantages?

Before you start: compressors

The compressor is the heart of the system. **Reciprocating** (piston), **rotary**, **scroll**, **screw** and **centrifugal** types exist. **Hermetic** = motor and compressor sealed together in one welded shell (no shaft seal, no leaks; household fridges and ACs).

Hermetic sealed compressor



Hermetic compressor: motor and compressor in one welded shell (course slides)

When the compressor and the motor run on the same shaft and are enclosed in a common welded casing, it is a **hermetic sealed compressor**. It removes the crankshaft seal that an open compressor needs to stop refrigerant leakage. It can be reciprocating or rotary, vertical or horizontal; used in household refrigerators and window/split ACs. Suction gas cools the motor windings.

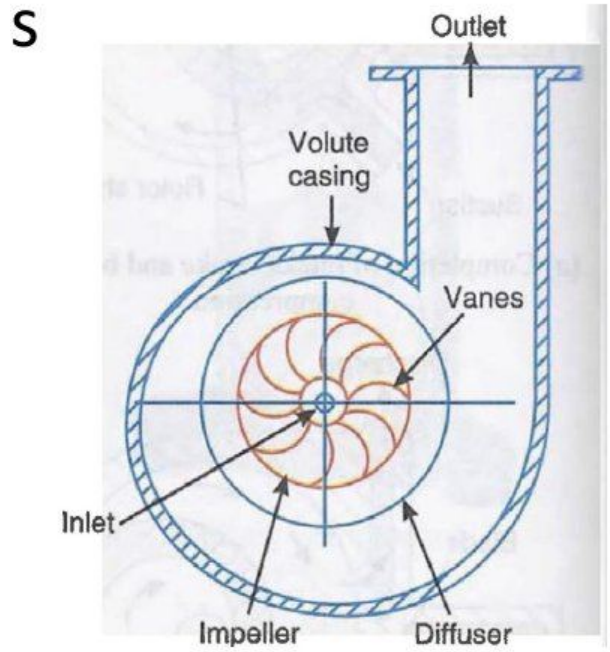
Advantages	Disadvantages
1. Leakage of refrigerant completely prevented (no shaft seal)	1. Maintenance is not easy; moving parts are inaccessible (shell must be cut)
2. Less noisy	2. A separate pump is needed for evacuation and charging
3. Small space; compact	3. Motor burnout contaminates the whole system
4. Simple lubrication; motor and compressor in sealed oil space	4. Speed cannot be changed; limited to small capacities
5. Cheaper for small units; no belt or coupling	5. Motor heat is added to the refrigerant

Items 1-4 in each column are from the slides; the rest are standard textbook points to add if more marks are needed.

Classification of compressors (2024 3c)

1. Method of compression: reciprocating, rotary, centrifugal
2. Working strokes: single acting, double acting
3. Number of stages: single stage, multi-stage
4. Method of drive: direct drive, belt drive
5. Location of prime mover: open, semi-hermetic, hermetic

Advantages of centrifugal over reciprocating



Centrifugal compressor: impeller, vanes, volute casing (course slides)

1. No valves, pistons, cylinders or connecting rods, so longer working life.
2. Little or no vibration, since there are no unbalanced masses.
3. Quiet and calm operation.
4. High speed (3000 rpm and above), so they couple directly to electric motors or steam turbines.
5. Handle large volumes of vapour.
6. Suited to 50-5000 TR (ton of refrigeration (3.5 kW)) systems and −90 °C to +10 °C.
7. High efficiency.
8. Large units need less floor area.

Disadvantages: surging below about 35% load, low pressure rise per stage, not practical below 50 TR, needs refrigerants with high specific volume.

Remember: Hermetic = one sealed shell: no leak, quiet, compact; but no access, no repair.

A6 Refrigerant selection: environmental and safety properties Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q3(b)** [10] What are the major criteria of refrigerant selection? Explain the environmental and safety properties of a good refrigerant.
- **2023 Q3(b)** [8] Discuss the environmental and safety properties to consider before selecting a refrigerant.
- **2021 Q2(b)** [8] Briefly discuss the desirable characteristics of a fluid to be used as a refrigerant.
- **2020 Q2(a)** [7] Explain the environmental and safety properties of refrigerant.

Before you start: what makes a good refrigerant

It must boil at a low temperature at a reasonable pressure, carry a lot of heat per kg, and be safe: not toxic, not flammable, and not harmful to the ozone layer (low ODP (ozone depletion potential)) or climate (low GWP (global warming potential)). Old CFCs like R12 damaged ozone; R22 (HCFC) is being phased out; R134a, R410A, R32, ammonia, CO_2 and hydrocarbons are used now.

What is a refrigerant

The working fluid that absorbs heat by evaporating at low temperature and pressure and rejects heat by condensing at higher temperature and pressure. Primary refrigerants (R134a, NH_3, R22) run the cycle; secondary refrigerants (brine, chilled water) carry cold from the plant to the load.

Major selection criteria (2024 3b)

1. Thermodynamic and thermo-physical properties
2. Environmental and safety properties
3. Economics (cheap, easily available)

Desirable thermodynamic properties (2021 2b)

Property	Should be	Reason
Suction (evaporator) pressure	Above atmospheric, high	No air/moisture ingress, easy leak detection, small compressor
Discharge pressure	Low	Light compressor and condenser
Pressure ratio	Small	High volumetric efficiency, low power
Latent heat of vaporization	Large	Small mass flow per kW
Isentropic index	Small	Low discharge temperature
Liquid specific heat	Small	More subcooling, less flash gas
Vapour specific heat	Large	Less superheating
Thermal conductivity	High	Better heat transfer
Viscosity	Low	Less pressure drop
Critical temperature / freezing point	Well outside working range	Condensation possible; no freezing

Environmental properties (2024, 2023, 2020)

- **ODP (ozone depletion potential)** should be zero (Montreal Protocol). Chlorine/bromine destroy ozone: CFCs (R11, R12) are phased out, HCFC (hydrochlorofluorocarbon) R22 is being phased out. Reference R11: ODP = 1.
- **GWP (global warming potential)**: climate impact of a gas compared with the same mass of CO_2 (GWP = 1). Should be as low as possible; zero-ODP but high-GWP fluids such as R134a will be regulated.
- **TEWI (total equivalent warming impact)** = direct effect (leakage, losses at recovery) + indirect effect (CO_2 from the electricity used). Low TEWI is preferred.

Safety properties

- **Toxicity:** should be non-toxic. Toxicity is relative: it matters only with the concentration and exposure time.
- **Flammability:** should be non-flammable and non-explosive; flammable ones (HCs, NH_3 partly) need special precautions. ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) classes: A/B = lower/higher toxicity, 1/2/3 = no/low/high flammability (R134a is A1, NH_3 is B2L).
- **Chemical stability** inside the system; **compatibility** with metals, gaskets and motor insulation.
- **Miscibility with oil:** NH_3 is not miscible (needs oil separator); R12 fully miscible; R22 partly (design for oil return).
- **Ease of leak detection.** High dielectric strength (hermetic motors).

Remember: Environment = ODP, GWP, TEWI. Safety = Toxic, Flammable, Stable, Compatible, Oil, Leak (TF-SCOL).

A13 Thermoelectric, magnetocaloric and other novel cooling

Good to have

Very likely · 90% Easy

PAST-PAPER QUESTIONS

- **2024 Q8(a)** [10] Explain the working principle of the thermoelectric cooling technique. How does reversing current affect its operation? (asked in Section B)
- **2024 Q8(b)** [10] Explain the reason why magneto-caloric cooling has not yet achieved commercial adoption in building HVAC (heating, ventilation and air-conditioning) system. (Section B)
- **2023 Q4(d)** [10] How thermo-electric cooling technique differs from solar-assisted cooling system? Explain which one is more efficient system and why?
- **2021 Q3(b)** [5] Differentiate between thermo-electric and electro-caloric cooling systems.

Before you start: novel cooling

Thermoelectric (Peltier): passing current through two different semiconductors moves heat from one junction to the other; no moving parts. **Magnetocaloric**: some materials heat up when magnetised and cool when demagnetised.

Not covered in the uploaded slides; written from standard textbooks (Arora, Khurmi). Check against class notes.

Thermoelectric cooling (2024 8a)

- Based on the **Peltier effect**: when DC current flows through a junction of two dissimilar semiconductors (n-type and p-type, usually bismuth telluride), heat is absorbed at one junction and released at the other.
- Many p-n couples are connected electrically in series and thermally in parallel between two ceramic plates: one plate gets cold (absorbs heat from the space), the other gets hot (rejects heat to a heat sink + fan).
- Heat pumped $Q_C = \alpha I T_C - \frac{1}{2} I^2 R - K\Delta T$.
- **Reversing the current** swaps the hot and cold sides: the device becomes a heater instead of a cooler. So one unit can heat or cool just by switching polarity.
- No moving parts or refrigerant, silent, compact, precise control; but low COP (0.3-0.7), used for small coolers, electronics, medical boxes.

Thermoelectric vs solar-assisted cooling (2023 4d)

Thermoelectric	Solar-assisted (absorption/adsorption or PV-driven VCR (vapour compression refrigeration))
Electric (DC), Peltier effect	Uses solar heat (collectors driving VAR (vapour absorption refrigeration)/adsorption) or solar electricity
COP (coefficient of performance) 0.3-0.7	Thermal COP 0.6-1.2 (VAR); PV + VCR overall COP high
Small capacity, no moving parts	Large capacity, needs collectors and storage

Solar-assisted cooling is more efficient for building loads: it uses free renewable heat and the absorption/VCR cycle has a much higher COP than Peltier modules, whose efficiency is limited by Joule heating and heat conduction back through the elements.

Thermoelectric vs electro-caloric (2021 3b)

Thermoelectric	Electro-caloric
Peltier effect: current through a junction moves heat	Applying an electric field to a dielectric (polymer/ceramic) orders its dipoles, so it heats up; removing the field cools it
Continuous DC current	Cyclic field with heat regenerator
Commercial, low COP	Research stage, potentially higher efficiency

Why magnetocaloric cooling is not commercial in buildings (2024 8b)

1. Needs strong magnetic fields (1-2 T) from rare-earth permanent magnets: heavy and expensive.
2. Good magnetocaloric materials (gadolinium alloys) are costly and scarce.
3. Small temperature change per cycle (2-5 K), so many regenerator stages and high cycle frequency are needed; capacity per kg is low.
4. Hysteresis, fatigue and corrosion of materials in water-based fluid.
5. Complex moving parts (rotating magnets), pumps and valves; poor heat transfer.
6. Vapour compression is cheap and mature, so there is no cost case yet.

Remember: Peltier: reverse current = swap hot/cold. Magnetocaloric: costly magnets, small ΔT, low power density.

A2 Effect of evaporator and condenser temperature on COP Must study

Very likely · 90% Easy

PAST-PAPER QUESTIONS

- **2024 Q2(a)** [12] Discuss the effect of evaporating and condensing temperature on the cycle efficiency with necessary diagrams.
- **2023 Q1(b)** [10] Explain the effects of evaporating and condensing temperature on the performance of vapor compression refrigeration cycle.
- **2021 Q1(b)** [12] How does the increase in condenser temperature affect on COP (coefficient of performance)? Also briefly explain the influence of evaporator temperature on COP. Which of the two temperatures have more influence on COP?

Before you start: why temperatures change COP

COP (coefficient of performance) = heat removed / work input. The compressor has to lift the heat from the evaporator temperature up to the condenser temperature. The bigger that lift, the more work per kJ of cooling, so COP falls. Ideal (Carnot) COP = $T_L/(T_H - T_L)$ in kelvin shows it.

(1) Effect of evaporating temperature (condensing temperature fixed)

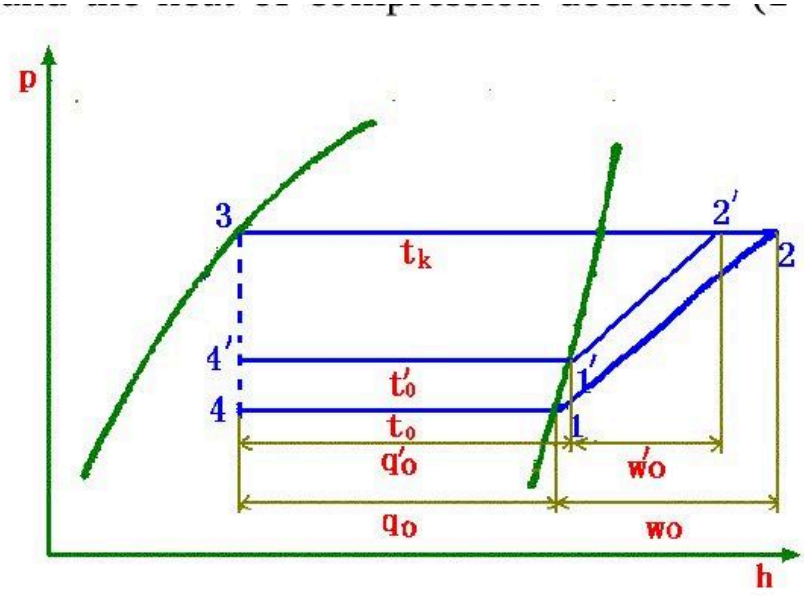
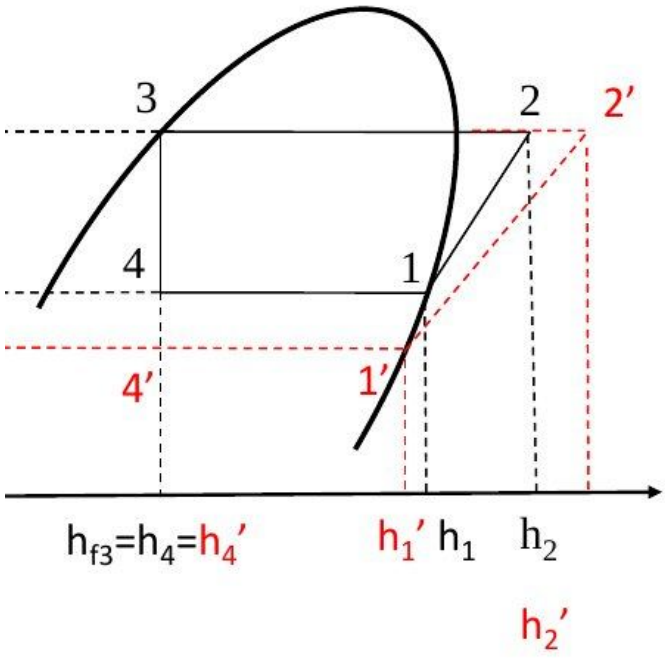


Fig.6.11.Effects of increased evaporating temperature

Increasing the evaporating temperature t_o to t_o' (course slides, Fig 6.11)

- **Refrigerating effect** $q_o = h_1 - h_3$ increases, because h_1 (saturated vapour enthalpy) rises.
- **Compressor work** $w = h_2 - h_1$ decreases, because the pressure lift is smaller.
- **Suction specific volume** v_1 falls a lot, so the volumetric refrigerating effect $(h_1 - h_3)/v_1$ and capacity rise quickly.
- So **COP increases** with higher evaporating temperature and falls sharply when it is lowered.

Lower suction pressure (from friction or low evaporator temperature) does the opposite: RE (refrigerating effect) falls from $(h_1 - h_4)$ to $(h_{1'} - h_{4'})$, work rises from $(h_2 - h_1)$ to $(h_{2'} - h_{1'})$, and COP decreases.



Effect of reduced suction (evaporator) pressure (course slides)

(2) Effect of condensing temperature (evaporating temperature fixed)

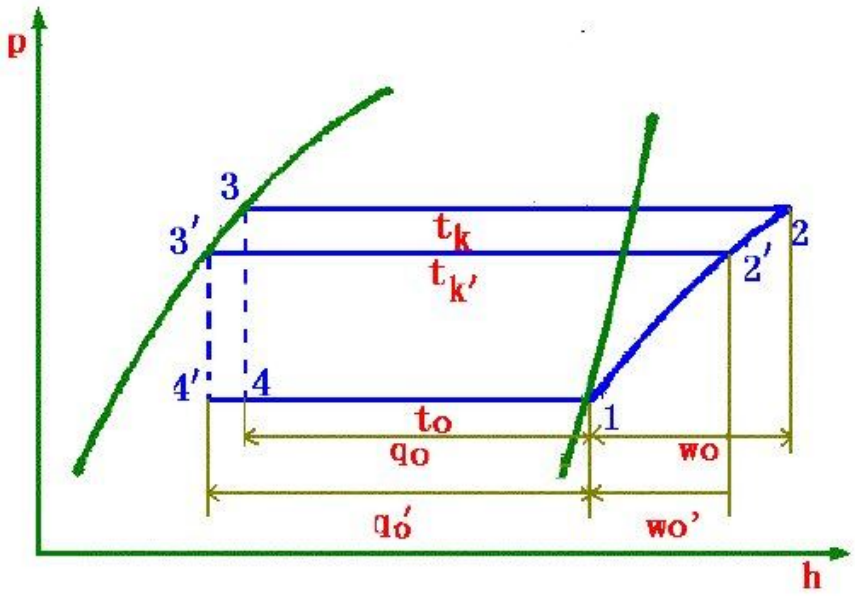
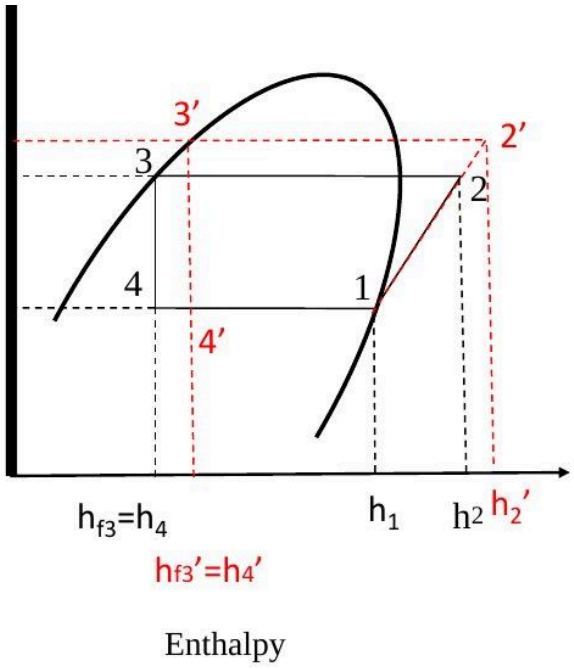


Fig.6.12.Effect of decreased condensing temperature

Decreasing the condensing temperature t_k to t_k' (course slides, Fig 6.12)

- Lower condensing temperature: h_3 drops, so **RE increases**; the pressure lift is smaller, so **work decreases**; heat rejected also decreases. **COP increases.**
- **Increase in condensing temperature** (the 2021 question) has the opposite effect: RE falls from $(h_1 - h_4)$ to $(h_1 - h_{4'})$, work rises from $(h_2 - h_1)$ to $(h_{2'} - h_1)$, discharge temperature rises, and **COP decreases**.



Effect of higher discharge (condenser) pressure (course slides)

Which one has more influence? (2021 1b)

The evaporator temperature has more influence on COP. From the Carnot COP = $T_e/(T_c - T_e)$:

$$\partial COP/\partial T_e = T_c/(T_c - T_e)^2 \quad > \quad |\partial COP/\partial T_c| = T_e/(T_c - T_e)^2 \quad \text{because } T_c > T_e$$

Raising the evaporator temperature also cuts v_1 sharply, so capacity improves too. Example with R22 (ideal cycle, −10 °C / 45 °C, COP 3.61): raising t_e by 5 K gives COP 4.13, while lowering t_c by 5 K gives 4.11.

Effect on cycle efficiency in one line (2024 2a)

The smaller the gap between condensing and evaporating temperatures, the higher the COP: keep the evaporator as warm and the condenser as cool as the application allows.

Remember: High evaporator, low condenser = high COP. Evaporator wins because $T_c > T_e$.

A1 Vapour compression refrigeration (VCR) cycle Must study Very likely · 80% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(b)** [12] Explain thermodynamic process of an ideal vapor compression refrigeration (VCR) cycle with necessary sketches.
- **2023 Q1(a)** [10] Describe the working principle of vapor compression refrigeration system with neat sketch.
- **2020 Q1(b)** [12] Discuss about the thermodynamic process of theoretical vapor compression refrigeration cycle with dry saturated vapor after compression with necessary diagrams.

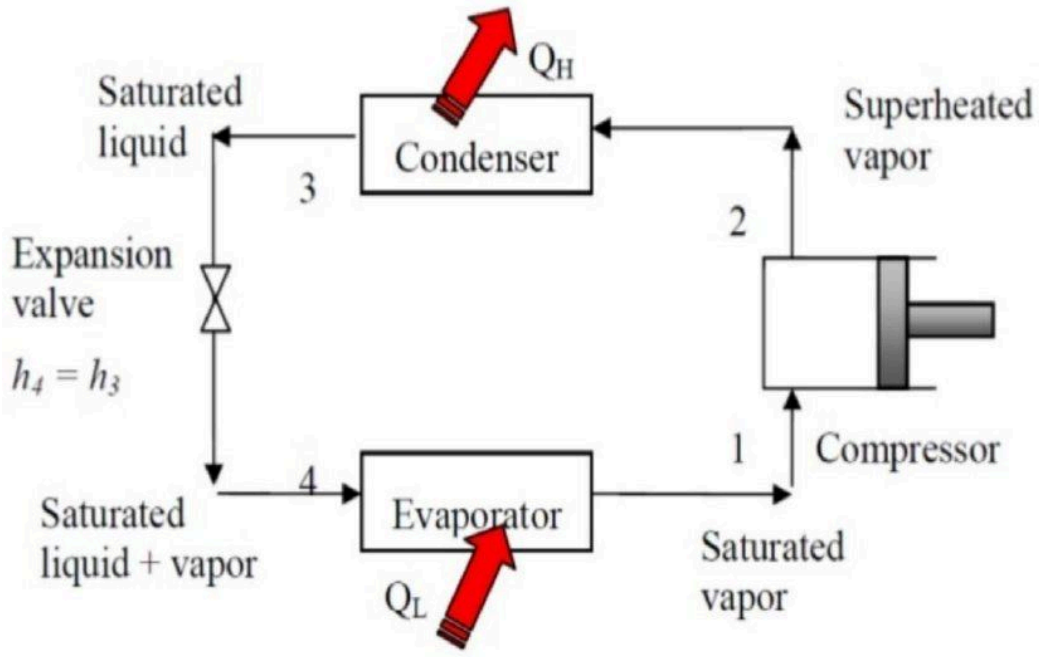
Before you start: how a fridge moves heat

Heat naturally flows from hot to cold. A refrigerator pushes it the other way using a liquid (refrigerant) that boils at a very low temperature. In the **evaporator** (inside the fridge) it boils and absorbs heat; the **compressor** squeezes the vapour so it becomes hot; in the **condenser** (outside, at the back) it gives that heat to the room and turns back to liquid; the **expansion valve** drops its pressure so it is cold again.

Enthalpy h (kJ/kg) = energy content per kg of refrigerant. On a p-h chart every process is a straight line, so all answers come from four h values.

Components and schematic

The VCR (vapour compression refrigeration) system has four components: **compressor, condenser, expansion valve, evaporator**. The refrigerant enters the compressor as saturated vapour, leaves the condenser as saturated liquid, is throttled to evaporator pressure, and evaporates by absorbing heat from the cold space.



Basic vapour compression refrigeration cycle (course slides)

The four processes (state points 1-2-3-4)

Process	Where	What happens	Energy per kg
1→2	Compressor	Isentropic compression of dry vapour from p_e to p_c; becomes superheated (or dry saturated)	$w = h_2 - h_1$
2→3	Condenser	Constant-pressure heat rejection: desuperheat then condense to saturated liquid	$q_c = h_2 - h_3$
3→4	Expansion valve	Throttling, no heat or work; pressure and temperature drop, part flashes to vapour	$h_4 = h_3$
4→1	Evaporator	Constant-pressure evaporation; absorbs heat from the space	RE (refrigerating effect) $= h_1 - h_4$

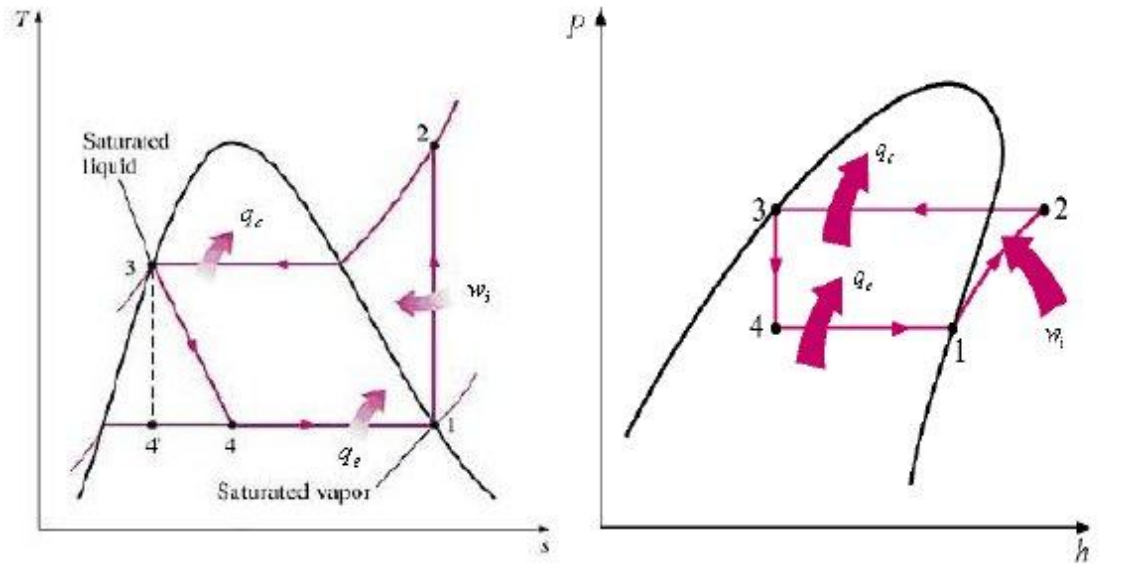


Fig 6.4.T-s and *p-h* Diagram for basic Vapor-Compression Refrigeration Cycle

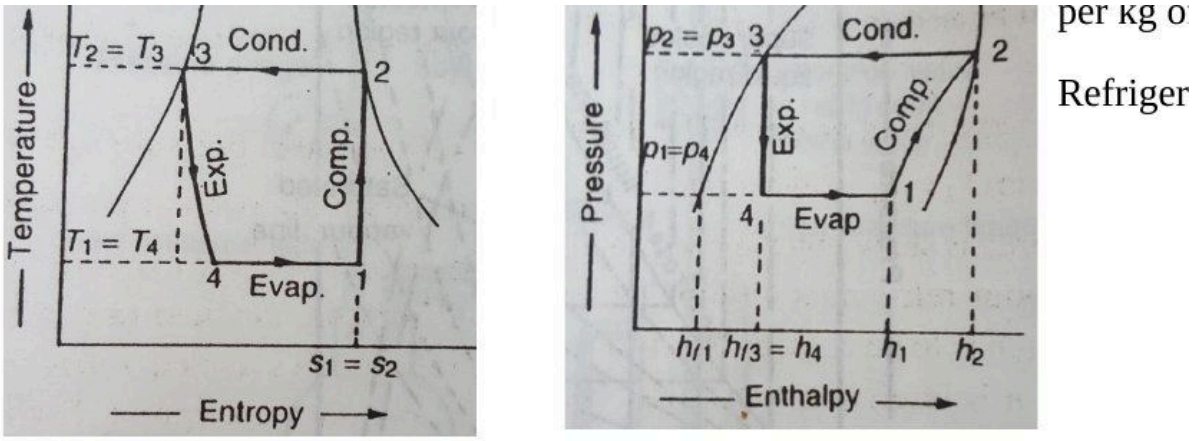
T-s and p-h diagrams of the ideal VCR cycle (course slides, Fig 6.4)

COP (coefficient of performance) = RE / w = $(h_1 - h_4) / (h_2 - h_1)$ $Q_e = m\,(h_1 - h_4)$ $P = m\,(h_2 - h_1)$

Why not the reversed Carnot cycle?

Carnot would need compression of a wet mixture (2 phases in the compressor) and expansion in a turbine of a very wet fluid. Both are impractical, so the ideal VCR cycle compresses dry vapour and uses a throttle valve instead of a turbine; the small work lost is not worth the cost of a turbine.

Cycle with dry saturated vapour after compression (2020 1b)



Dry saturated vapour after compression: T-s and p-h (course slides)

Here state 1 is wet (dryness x_1) so that isentropic compression ends exactly on the saturated vapour line at state 2 ($s_1 = s_2 = s_{g2}$). Then condensation 2→3 is only latent heat at constant temperature $T_2 = T_3$.

$h_1 = h_{f1} + x_1\, h_{fg1}$, with x_1 from $s_1 = s_{f1} + x_1(s_{g1} - s_{f1}) = s_{g2}$; $COP = (h_1 - h_{f3})/(h_2 - h_1)$

Subcooling and superheating (often asked with the cycle)

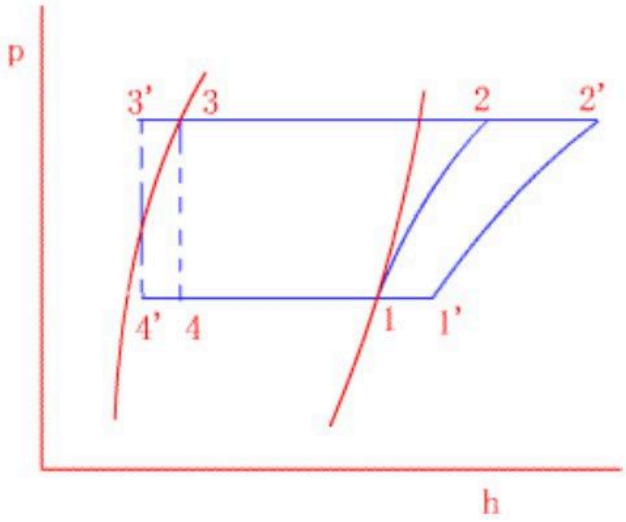


Fig.6.7. Subcooling of the liquid and superheating of the vapor

Subcooling (3 to 3′) and superheating (1 to 1′) on p-h (course slides, Fig 6.7)

- **Subcooling** cools the liquid below saturation (3→3′). RE rises by $h_3 - h_{3'}$, work is unchanged, so COP always rises.
- **Superheating** (1→1′) increases both RE and work; COP may rise, fall or stay the same. Some superheat protects the compressor from liquid; too much overheats oil.

Remember: C-C-E-E: Compress (w = h2−h1), Condense, Expand (h4 = h3), Evaporate (RE = h1−h4).

A8 Vapour absorption (VAR): single, double, triple effect Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q4(b)** [13] Describe the commercial double-effect VAR (vapour absorption refrigeration) system with necessary sketch.
- **2024 Q4(c)** [8] Briefly explain the different heat sources of VAR system.
- **2023 Q3(a)** [13] Write down the classification of vapor absorption refrigeration system. Describe the triple effect vapor absorption system with a schematic diagram.
- **2023 Q3(d)** [6] Differentiate between VCR (vapour compression refrigeration) and VAR systems.
- **2021 Q4(d)** [10] Draw the schematic of a single effect absorption refrigeration system. Also explain how it is different from double effect absorption refrigeration system.
- **2021 Q4(b)** [6] What are the advantages of vapor absorption system over vapor compression refrigeration system?
- **2020 Q3(a)** [10] Draw double effect absorption refrigeration system and explain how it is different from triple effect absorption refrigeration system.

Before you start: absorption refrigeration

Instead of a compressor, an absorption system uses **heat** (steam, hot water, gas flame, waste heat). The refrigerant vapour is absorbed into a liquid (water absorbs ammonia; lithium bromide solution absorbs water vapour), pumped up as liquid (very little work), then boiled out again in a **generator** by heat. Double and triple effect reuse the generator heat in more stages for higher COP (coefficient of performance).

Basic (single-effect) vapour absorption system (2021 4d)

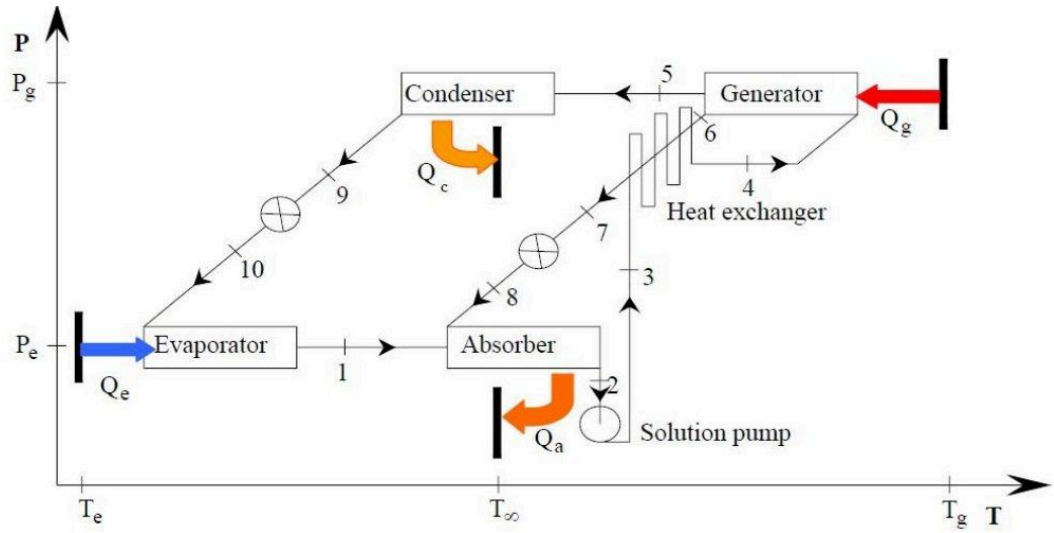


Fig.14.7: Basic vapour absorption refrigeration system with a solution heat exchanger on a pressure vs temperature diagram

Basic vapour absorption system with solution heat exchanger on p-T diagram (course slides)

The compressor of a VCR is replaced by a **thermal compressor**: absorber + solution pump + generator. Input is heat, not work.

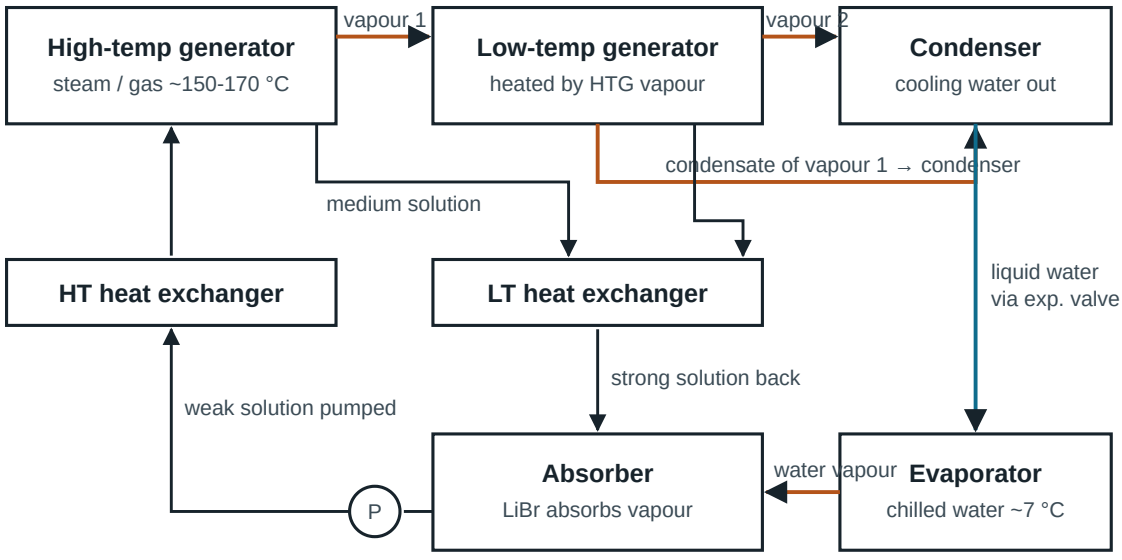
1. **Evaporator** (low pressure): refrigerant evaporates and produces cooling Q_e.
2. **Absorber**: the vapour is absorbed by a strong solution; heat Q_a is removed by cooling water. The weak solution is formed.
3. **Solution pump** raises the liquid to generator pressure (very little work, since it pumps liquid).
4. **Heat exchanger**: hot strong solution returning to the absorber preheats the weak solution.
5. **Generator**: heat Q_g drives the refrigerant vapour out of the solution; the strong solution returns through the HX and a valve to the absorber.
6. **Condenser**: vapour condenses (Q_c) and goes through the expansion valve to the evaporator.

Working pairs: **H_2O-LiBr** (water refrigerant, LiBr (lithium bromide) absorbent; above 0 °C, air conditioning) and **NH_3-H_2O** (ammonia refrigerant; below 0 °C, refrigeration). $COP_{VAR} = Q_e/(Q_g + W_{pump})$, about 0.7 for single effect.

Classification of VAR systems (2023 3a)

- By working pair: water-LiBr, ammonia-water (and Electrolux NH_3-H_2O-H_2, pump-free)
- By number of effects: half, single, double, triple effect
- By heat input: indirect fired (steam / hot water), direct fired (gas, oil), waste heat, solar
- By number of shells: single shell, two shell

Double-effect system (2024 4b, 2020 3a, 2021 4d)



Double-effect LiBr-water system (series flow), drawn for these notes. Vapour 1 from the HTG is the heat source of the LTG.

- Has **two generators**: a high-temperature generator (HTG) heated by steam (about 8 bar) or by a burner, and a low-temperature generator (LTG).
- Vapour from the HTG is not sent straight to the condenser; it **condenses inside the tubes of the LTG** and its latent heat boils more refrigerant out of the solution there. The same heat input is used twice.
- Both vapours end up in the condenser, then expansion valve, evaporator and absorber as in single effect. Two solution heat exchangers (HT and LT) recover heat.
- COP ≈ **1.0-1.2** (single effect ≈ 0.7), so steam consumption is about half. Needs higher source temperature (≈150-170 °C).

Single vs double effect

	Single effect	Double effect
Generators	One	Two (HTG + LTG)
Heat source	80-120 °C hot water / low-pressure steam / waste heat	150-170 °C steam or direct fired
COP	≈0.7	≈1.0-1.2
Cost / complexity	Lower	Higher; more corrosion risk

Triple-effect system (2023 3a, 2020 3a)

Adds a **third, highest-temperature generator**: vapour from generator 1 heats generator 2, and vapour from generator 2 heats generator 3, so the heat input is used three times. COP ≈ **1.4-1.7**, but it needs about 200-230 °C heat (direct fired), runs at higher pressure, and has severe corrosion and crystallization limits, so it is mostly direct-fired and expensive.

Double effect	Triple effect
2 generators, heat used twice	3 generators, heat used three times
COP ≈ 1.2	COP ≈ 1.5-1.7
Source ≈150-170 °C	Source ≈200-230 °C
Mature, common commercial chiller	Costly, corrosion problems, few units

The double and triple effect details are not in the uploaded slides; they are from standard textbooks (Arora, ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers)). Check against your class notes if they differ.

VCR vs VAR (2023 3d)

Compression systems	Absorption systems
Work operated	Heat operated
High COP	Low COP (currently maximum ≈ 1.4)
Performance (COP and capacity) very sensitive to evaporator temperatures	Performance not very sensitive to evaporator temperatures
System COP reduces considerably at part loads	COP does not reduce significantly with load
Liquid at the exit of evaporator may damage compressor	Presence of liquid at evaporator exit is not a serious problem
Performance is sensitive to evaporator superheat	Evaporator superheat is not very important
Many moving parts	Very few moving parts
Regular maintenance required	Very low maintenance required
Higher noise and vibration	Less noise and vibration
Small systems are compact and large systems are bulky	Small systems are bulky and large systems are compact
Economical when electricity is available	Economical where low-cost fuels or waste heat is available

Comparison between compression and absorption systems (course slides)

Advantages of VAR over VCR (2021 4b)

1. Runs on low-grade heat (waste heat, solar, steam), so lower operating cost where heat is cheap.
2. Few moving parts (only a pump): quiet, less vibration and maintenance.
3. Uses natural refrigerants (water, ammonia); no CFCs.

4. Performance hardly falls at part load; COP not sensitive to evaporator temperature.
5. Liquid at evaporator exit is not a problem (no compressor to damage).
6. Large capacities are economical; useful for heat recovery.

Heat sources of VAR (2024 4c)

- Steam (low pressure for single effect, ~8 bar for double effect)
- Hot water from boilers or district heating
- Direct firing of natural gas or oil
- Waste heat: engine jacket water, exhaust gases, process heat, cogeneration
- Solar thermal collectors; geothermal water

Remember: VAR = heat in, pump instead of compressor. Each extra generator reuses the heat: COP 0.7 → 1.2 → 1.6.

A5 Multistage / multipressure systems and flash cooler Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q3(a)** [13] Why and when multi-stage VCR (vapour compression refrigeration) system is required? Explain two-stage expansion system with a flash cooler with necessary sketch.
- **2023 Q2(a)** [10] Why multi-stage vapor compression is required? Write down the advantages and disadvantages of a multi-stage vapor compression system.
- **2023 Q2(b)** [13] Describe two-stage expansion system with a flash cooler with necessary diagrams.
- **2021 Q2(a)** [12] What is meant by multipressure systems? Discuss about the individual compressors-multiple expansion valves with the required diagrams.
- **2020 Q2(b)** [11] Define multistage compression. Explain the single compressor-multiple expansion valves with the required p-h diagram.

Before you start: why multistage

If the evaporator is very cold and the condenser hot, one compressor has to work across a huge pressure ratio: it gets hot, inefficient, and much of the liquid flashes to vapour in the expansion valve. Splitting compression into two stages with an intercooler and a **flash chamber** (which removes the flash vapour at intermediate pressure) saves work.

Why and when multistage is needed

A single stage works well when the pressure ratio between condenser and evaporator is moderate. A limit of about **40 K** between condensing and evaporating temperature is recommended (Gosney); above that, use multistage (or cascade) systems. With a large pressure ratio (very low evaporator or high condenser temperature), a single stage suffers:

1. Cylinder size becomes too large.
2. Discharge temperature becomes very high; it may heat the cylinder head or burn the lubricating oil.
3. Heat cannot be rejected in the short compression time.
4. Friction losses and running cost are high.
5. Volumetric efficiency is low.

Multipressure system: a system with two or more low-side pressures (several evaporators at different temperatures) or with compression in stages, giving more than one pressure level between p_e and p_c.

Advantages of multistage compression

1. Less work per kg (with intercooling) for the same delivery pressure.
2. Lower compression ratio per stage, so better volumetric and compressor efficiency.
3. Lower discharge temperature; better lubrication.
4. Liquid enters the evaporator at lower enthalpy, so higher refrigerating effect.
5. Cylinder sizes can suit the volume of refrigerant; less leakage loss.
6. More uniform torque, smaller flywheel; handles load variation.

Disadvantages

1. Higher first cost: extra compressor, intercooler, flash chamber, controls.
2. More complex piping and control, more maintenance.
3. Not economical when the temperature lift is small.

Interstage pressure for equal pressure ratios: $p_i = \sqrt{(p_c\ p_e)}$

Two-stage expansion with a flash cooler (2024 3a, 2023 2b)

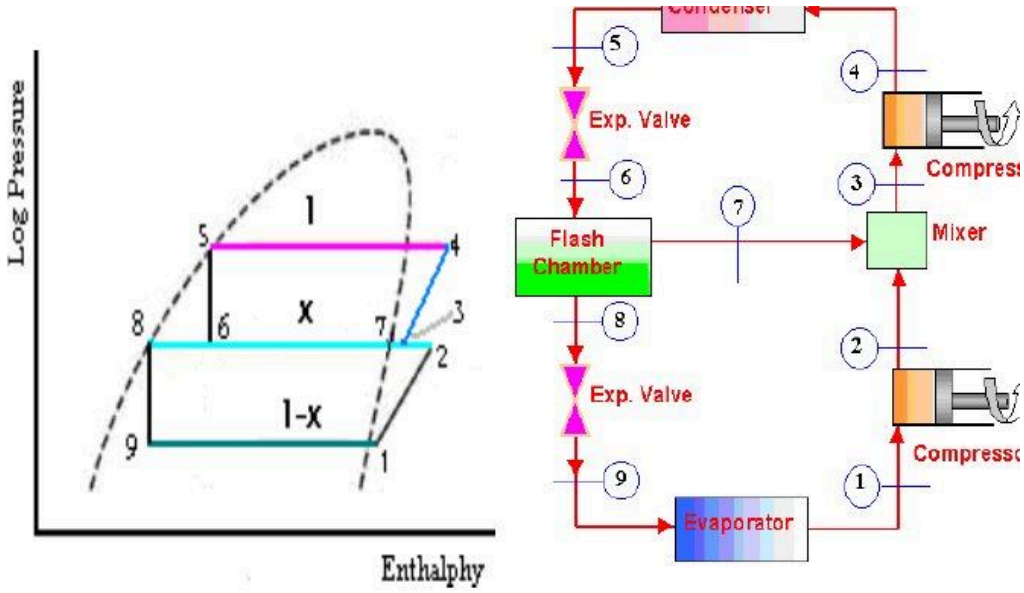


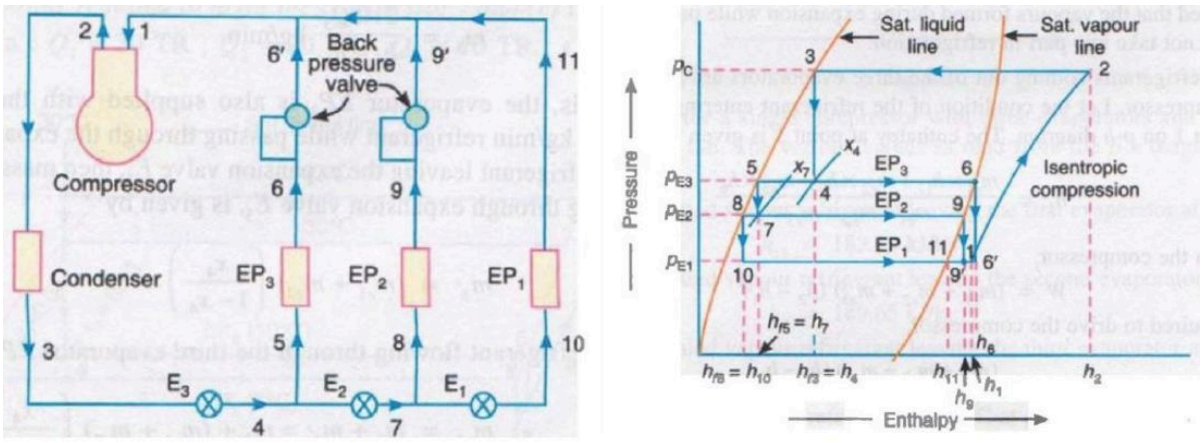
Fig.6.13 Two stage expansion system with a flash cooler

Two-stage expansion system with a flash cooler (course slides, Fig 6.13)

1. Dry saturated vapour at 1 enters the **low-stage compressor** and is compressed to interstage pressure p_i (point 2).
2. It mixes with flash vapour from the flash cooler (economizer); the mixture at 3 enters the **high-stage compressor** and leaves at p_c (point 4).
3. Condenser: desuperheated and condensed to liquid (point 5).
4. First expansion valve (float valve) throttles 5→6 into the flash cooler. A part x flashes to vapour (7); its latent heat cools the remaining liquid to saturation at p_i (8).
5. Flash vapour (7) goes to the high-stage compressor; liquid (8) goes through the second expansion valve to 9 and into the evaporator.
6. Evaporator: liquid evaporates to 1, completing the cycle.

Benefit: flash gas is not passed through the evaporator or low-stage compressor, the liquid entering the evaporator has lower enthalpy (more RE (refrigerating effect)), and the high-stage suction is cooler, so total work is less and COP (coefficient of performance) is higher.

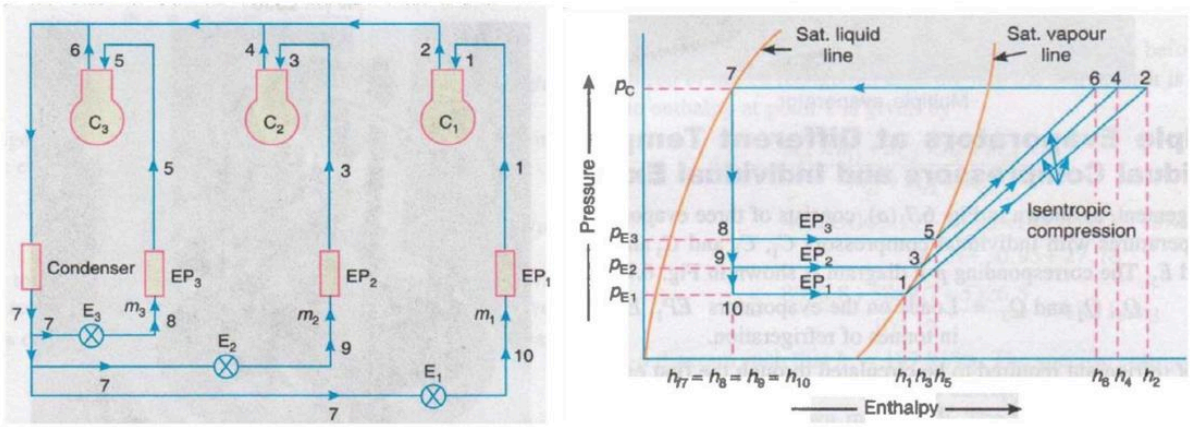
Single compressor with multiple expansion valves (2020 2b)



Three evaporators at different temperatures, single compressor, expansion valves in series E3, E2, E1 with back-pressure valves (course slides)

- Liquid from the condenser (3) is expanded in steps: E_3 to p_{E3}, the vapour formed is separated and only liquid continues to E_2, then E_1. Each evaporator gets liquid only, so the flash gas does no useless work in the colder evaporators.
- Back-pressure valves on EP_3 and EP_2 drop their outlet pressure to the lowest pressure p_{E1}; all vapours mix (state 1) and enter one compressor.
- Mixed enthalpy $h_1 = (m_1h_{11} + m_2h_{9'} + m_3h_{6'})/(m_1 + m_2 + m_3)$; $W = (m_1 + m_2 + m_3)(h_2 - h_1)$; $COP = 210(Q_1 + Q_2 + Q_3)/W$

Individual compressors with multiple expansion valves (2021 2a)



Multiple evaporators with individual compressors C1, C2, C3 (course slides). With multiple expansion valves, the valves E3, E2, E1 are connected in series as in the figure above.

- Each evaporator has its **own compressor** that takes its vapour straight to condenser pressure, so no back-pressure valve wastes pressure; each evaporator works at its own pressure.
- Liquid is expanded in series ($E_3 \rightarrow E_2 \rightarrow E_1$); the flash vapour formed at each stage is drawn off by that stage's compressor, and only liquid passes to the next, colder evaporator.
- $m_1 = 210Q_1/(h_1 - h_f)$ etc.; total power $P = P_1 + P_2 + P_3$, with $P_i = m_i(h_{out} - h_{in})/60$ kW.
- $COP = 210(Q_1 + Q_2 + Q_3) / [m_1(h_2 - h_1) + m_2(h_4 - h_3) + m_3(h_6 - h_5)]$; it is higher than with one compressor and back-pressure valves.

Remember: >40 K lift → go multistage. Flash cooler = flash gas skips the evaporator and low stage. $p_i = \sqrt{(p_c p_e)}$.

A10 Performance of a practical plant: EER, SEER, kW/ton Good to have

Likely · 70% Easy

PAST-PAPER QUESTIONS

- **2024 Q2(b)** [8] How the full-load and part-load performance of a practical refrigeration plant measured? Briefly explain.
- **2024 Q2(c)** [15] Write notes on: (i) EER (energy efficiency ratio), (ii) SEER (seasonal energy efficiency ratio), (iii) kW/ton, (iv) Energy Star, (v) HSPF.
- **2023 Q1(c)** [8] How the performance of a practical refrigeration is evaluated? Describe.

Before you start: efficiency ratings

EER = cooling output in Btu/h per watt of electricity (EER = 3.412 × COP (coefficient of performance)). **SEER** = the same averaged over a cooling season. **kW/ton** = electric kW per TR (ton of refrigeration (3.5 kW)) of cooling (lower is better).

Measuring performance of a practical plant (2024 2b, 2023 1c)

COP of a real plant depends on outdoor conditions, components, application and season, so standard indices are used:

- **Full-load performance:** COP, EER, or kW per ton at rated (100% load, standard test) conditions.
- **Part-load performance:** SEER for small/unitary units; **IPLV** (integrated part-load value, at ARI standard conditions) and NPLV (non-standard) for chillers.

```
IPLV (or NPLV) = 0.01A + 0.42B + 0.45C + 0.12D
```

A, B, C, D = efficiency at 100% load (29.4 °C condenser water), 75% (23.9 °C), 50% (18.3 °C) and 25% (18.3 °C). The weights show that plants run mostly at 50-75% load.

Short notes (2024 2c)

Index	Meaning	Formula / unit
EER	Energy efficiency ratio: cooling output per unit electrical input at one steady rated condition (outdoor 35 °C, indoor 27 °C)	EER = Btu/h ÷ W; COP = EER/3.412
SEER	Seasonal EER: total cooling over a whole season (Btu) ÷ total energy used (Wh); includes part-load running. US minimum 13 since 2006	Btu/Wh
kW/ton	Electrical input per ton of cooling; used for large chillers. Lower is better (good chillers 0.5-0.6)	kW/ton = 12/EER = 3.517/COP
Energy Star	US EPA label for products with high efficiency (above minimum standards); helps buyers pick efficient units (e.g. split ≥ 14.5 SEER, 12 EER, 8.2 HSPF)	Label
HSPF	Heating seasonal performance factor: heat output over the heating season (Btu) ÷ electricity used (Wh), for air-source heat pumps	Btu/Wh

Remember: Full load: COP, EER, kW/ton. Part load: SEER, IPLV = 0.01A + 0.42B + 0.45C + 0.12D. kW/ton = 12/EER.

A15 Cooling towers and condensers Skip if short Likely · 70% Easy

PAST-PAPER QUESTIONS

- **2024 Q4(d)** [6] Write down the classifications of cooling towers.
- **2023 Q8(c)** [9] Write down the classifications of cooling towers? Also describe how the performance of cooling tower is evaluated? (Section B)
- **2023 Q4(c)** [5] Write down the differences between air cooled condenser and water cooled condenser.

Before you start: rejecting heat

All heat removed plus compressor work must be rejected in the condenser: air-cooled (fins + fan), water-cooled (shell and tube, with a cooling tower to cool the water) or evaporative. A cooling tower cools water by evaporating a small part of it into air.

Not covered in the uploaded slides; written from standard textbooks (Arora, Khurmi). Check against class notes.

Classification of cooling towers (2024 4d, 2023 8c)

- By draft: **natural draft** (atmospheric spray, hyperbolic chimney) and **mechanical draft** (forced draft: fan at inlet pushes air; induced draft: fan at top pulls air).
- By air-water flow: **counter-flow** (air up, water down) and **cross-flow** (air horizontal).
- By construction: open (wet) and closed-circuit (dry/fluid coolers); field-erected and factory-assembled.

Performance of a cooling tower

$\text{Range} = T_{w,in} - T_{w,out}$; $\text{Approach} = T_{w,out} - T_{wb,air}$; $\text{Effectiveness} = \text{Range}/(\text{Range} + \text{Approach})$

Also: cooling capacity = $m_w c_p \times$ range; L/G (water/air) ratio; evaporation loss ≈ $0.00085 \times m_w \times$ range (°F) or about 1% per 6 K; drift and blowdown; cycles of concentration.

Air-cooled vs water-cooled condenser (2023 4c)

Air-cooled	Water-cooled
Rejects heat to air by fans	Rejects heat to water (cooling tower loop)
Higher condensing temperature, lower COP (coefficient of performance)	Lower condensing temperature, higher COP
No water needed, simple, low maintenance	Needs water, tower, pumps, water treatment
Small/medium units (<~100 TR (ton of refrigeration (3.5 kW)))	Large plants
Bigger, noisy fans, outdoor placement	Compact condenser; scaling and fouling problems

Remember: Range = in − out. Approach = out − WBT (wet bulb temperature). Effectiveness = R/(R + A).

A12 Air refrigeration and aircraft cooling Good to have Likely · 60% Medium

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [10] Describe the working principle of an air refrigerator system. Also mention the necessity of cooling for an aircraft.
- **2021 Q3(a)** [10] What are the different methods of air refrigeration systems? Explain the simple air cooling system with necessary diagrams.
- **2020 Q3(c)** [13] In aircraft refrigeration unit using air cycle, 50 kg/min of air at 200 cm Hg gauge and 205 °C are bled off the air compressor serving the jet engine. The bled air passes through a heat exchanger leaving at 180 cm Hg gauge and 80 °C. It is then expanded through a small cooling turbine to 20 cm Hg vacuum and −10 °C. The air exhausted out of the plane is at 25 °C. Assume c_p = 1.0 kJ/kgK. Determine the cooling capacity in TR (ton of refrigeration (3.5 kW)), and COP (coefficient of performance) based on input power to bled-off air.

Before you start: air refrigeration

Air itself can be the refrigerant: compress it, cool it at high pressure, then expand it in a turbine so it becomes very cold (reversed Brayton cycle). COP is low, but air is free, light and safe, so it is used in aircraft cabins.

Not covered in the uploaded slides; written from standard textbooks (Arora, Khurmi). Check against class notes.

Methods of air refrigeration (2021 3a)

1. Simple air cooling system
2. Simple evaporative system
3. Boot-strap system
4. Boot-strap evaporative system
5. Reduced ambient system
6. Regenerative air cooling system

Air refrigerator (Bell-Coleman / reversed Brayton) working principle (2023 4a)

Air is the refrigerant and stays a gas. Air from the cold room is compressed isentropically (1-2), cooled at constant pressure in a heat exchanger by cooling water (2-3), expanded isentropically in an expander/turbine to a very low temperature (3-4), and then absorbs heat from the cold room at constant pressure (4-1).

$$COP = (T_1 - T_4)/[(T_2 - T_3) - (T_1 - T_4)] = 1/(r_p^{(\gamma-1)/\gamma} - 1)$$

Simple air cooling system for aircraft

Ram air is compressed by the engine compressor; part is bled off, cooled in a heat exchanger by ram air, expanded in a cooling turbine to low temperature and supplied to the cabin. The turbine drives the cooling fan of the heat exchanger.

Why aircraft need cooling

- Heat from passengers, avionics and equipment, solar radiation through the fuselage.
- Ram (kinetic) heating of the skin at high speed; air from the compressor is very hot.
- Cabin must be pressurized and ventilated with fresh air; outside air at altitude is thin.

- Air cycle is chosen because it is light, uses no toxic refrigerant, and air is free and also gives pressurization.

Numerical (2020 3c)

> m = 50 kg/min, c_p = 1.0; turbine exit −10 °C; cabin exhaust 25 °C; bleed 205 °C.
> Refrigerating effect = m c_p(T_{cabin} − $T_{turbine\ exit}$) = 50 × 1.0 × (25 − (−10)) = 1750 kJ/min
> **Capacity = 1750/210 = 8.33 TR**
> Input energy carried by bled air (relative to the exhaust state) = m c_p(205 − 25) = 50 × 180 = 9000 kJ/min
> **COP = 1750/9000 = 0.19**

The paper does not say what "input power to bled-off air" is measured from; taking the cabin exhaust state (25 °C) as reference is the usual textbook choice. State it in the answer.

Remember: Air cycle: compress, cool, expand in turbine, absorb. Aircraft TR = m c_p(T_{cabin} − $T_{turbine}$)/210.

A9 LiBr-water practical problems (crystallisation) Good to have Likely · 60% Medium

PAST-PAPER QUESTIONS

- **2024 Q4(a)** [8] Briefly explain the various practical problems in water-lithium bromide vapor absorption refrigeration (VAR) system.
- **2021 Q4(c)** [7] What are the practical problems in LiBr-water vapor absorption system? Write a short note on crystallization problem.

Before you start: crystallisation

Lithium bromide solution is like very salty water. If it becomes too strong or too cold, salt crystals form and block the pipes (like sugar syrup crystallising). Plants avoid this with controls on concentration and temperature.

Not covered in the uploaded slides; written from standard textbooks (Arora, Khurmi). Check against class notes.

Practical problems in LiBr-water systems (2024 4a, 2021 4c)

1. **Crystallization:** if the strong solution gets too concentrated or too cold (e.g. in the solution heat exchanger at low load, low cooling-water temperature, or power failure), LiBr (lithium bromide) salt crystallizes and blocks the pipes. Prevented by limiting concentration (<65%), controlling cooling water temperature, dilution cycle at shutdown, and an anti-crystallization (bypass) pipe.
2. **Very low (vacuum) pressure:** water refrigerant runs at about 0.9 kPa in the evaporator, so air leaks in; purge units are needed to remove non-condensables, which otherwise cut capacity.
3. **Freezing of water:** cannot be used below about 4 °C evaporator temperature since water freezes at 0 °C.
4. **Corrosion:** LiBr is corrosive to steel and copper with oxygen present; inhibitors (lithium chromate, molybdate) are added.
5. **Large specific volume of water vapour:** needs large vessels and pipes; pressure drop must be tiny, so components are housed in one or two shells.
6. **Hydrostatic head:** even small liquid heights change the boiling pressure, so sprays and falling films are used.

Crystallization note: crystallization happens when the solution's state crosses the crystallization line on the Dühring/PTX chart. It is most likely at the hot-side outlet of the solution heat exchanger where the strong solution is coldest.

Remember: LiBr problems: Crystallization, Air leak (vacuum), Corrosion, Freezing (CACF).

A4 R22 VCR cycle numerical (with table / p-h chart) Must study Likely · 70% Hard

PAST-PAPER QUESTIONS

- **2024 Q1(c)** [13] A refrigeration system using R22 is to have a refrigerating capacity of 80 kW. The system follows VCR (vapour compression refrigeration) cycle with 5 °C subcooling and 10 °C superheating in which the evaporating and condensing temperature are −10 °C and 45 °C, respectively. Determine: (i) the volume flow rate of refrigerant, (ii) COP (coefficient of performance), (iii) the power required to drive the compressor.
- **2021 Q1(c)** [15] Same question as 2024 1(c): R22, 80 kW, 5 °C subcooling, 10 °C superheating, evaporating −10 °C, condensing 45 °C. Find volume flow rate, COP and compressor power.
- **2020 Q2(c)** [12] A standard VCR system developing 50 kW of refrigeration using R22 refrigerant operates with a condensing temperature of 35 °C and an evaporating temperature of −10 °C. The enthalpy of the refrigerant after the compression is 435.2 kJ/kg. Using the table, determine: (i) the refrigeration effect in kJ/kg; (ii) the circulation rate of refrigerant in kg/s; (iii) the power required by the compressor in kW; and (iv) COP.

Table: Properties of saturated liquid and saturated vapor of R22

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Entropy (kJ/kgK)	
		Liquid	Vapor	Liquid	Vapor
−12	3.3040	186.09	400.27	0.9484	1.7686
−8	3.8059	190.69	401.85	0.9657	1.7621
34	13.215	241.84	414.79	1.1419	1.7050
36	13.898	244.44	415.18	1.1501	1.7024

Before you start: reading refrigerant tables

State 1 = vapour leaving the evaporator (to compressor); 2 = hot vapour leaving the compressor; 3 = liquid leaving the condenser; 4 = after the expansion valve (h4 = h3 because throttling does no work and loses no heat).

Superheat = vapour heated above its boiling temperature; **subcooling** = liquid cooled below its condensing temperature. Isentropic compression means s2 = s1 (read h2 from the chart along the constant-entropy line).

Problem (2024 1c and 2021 1c): R22, 80 kW, t_e = −10 °C, t_c = 45 °C, 5 K subcooling, 10 K superheat

Property values below are R22 table / p-h chart values (IIR reference, h = 200 kJ/kg at 0 °C liquid). Readings from your chart may differ by 1-2%; the method is what earns the marks.

Step 1: state points

p_e = 3.55 bar (sat. at −10 °C); p_c = 17.29 bar (sat. at 45 °C)

State 1: superheated vapour at 3.55 bar, −10 + 10 = 0 °C: h_1 = 408.2 kJ/kg, s_1 = 1.792 kJ/kg K, v_1 = 0.0686 m³/kg

State 2: isentropic to 17.29 bar ($s_2 = s_1$): h_2 = 450.4 kJ/kg ($t_2 \approx 81$ °C)

State 3: liquid at 45 − 5 = 40 °C: $h_3 = h_f$(40 °C) = 249.6 kJ/kg; $h_4 = h_3$ = 249.6 kJ/kg

Step 2: per kg

RE (refrigerating effect) = $h_1 - h_4$ = 408.2 − 249.6 = 158.6 kJ/kg

w = $h_2 - h_1$ = 450.4 − 408.2 = 42.2 kJ/kg

Step 3: mass flow m = 80 / 158.6 = 0.5045 kg/s

(i) Volume flow at suction V = m v_1 = 0.5045 × 0.0686 = 0.0346 m³/s (2.08 m³/min)

(ii) COP = 158.6 / 42.2 = 3.75

(iii) Power = m w = 0.5045 × 42.2 = 21.3 kW

On the p-h sketch show: the horizontal evaporator line at 3.55 bar extending past the saturated vapour line to 1 (superheat), the isentropic line 1→2, the condenser line at 17.29 bar ending left of the saturated liquid line at 3 (subcooled), and the vertical throttling line 3→4. Use Fig 6.7 under A1 as the template.

Problem (2020 2c): R22, 50 kW, t_c = 35 °C, t_e = −10 °C, h_2 = 435.2 kJ/kg

Table: Properties of saturated liquid and saturated vapor of R22

Temperature (°C)	Pressure (bar)	Enthalpy (kJ/kg)		Entropy (kJ/kgK)	
		Liquid	Vapor	Liquid	Vapor
−12	3.3040	186.09	400.27	0.9484	1.7686
−8	3.8059	190.69	401.85	0.9657	1.7621
34	13.215	241.84	414.79	1.1419	1.7050
36	13.898	244.44	415.18	1.1501	1.7024

Table given with the 2020 question

−10 °C lies midway between −12 and −8: $h_1 = h_g$ = (400.27 + 401.85)/2 = 401.06 kJ/kg

35 °C lies midway between 34 and 36: $h_3 = h_f$ = (241.84 + 244.44)/2 = 243.14 kJ/kg = h_4

(i) RE = 401.06 − 243.14 = 157.92 kJ/kg

(ii) m = 50 / 157.92 = 0.3166 kg/s

(iii) P = m ($h_2 - h_1$) = 0.3166 × (435.2 − 401.06) = 0.3166 × 34.14 = 10.81 kW

(iv) COP = 157.92 / 34.14 = 4.63

Remember: Find h1, h2, h3 (= h4) and v1 first. Then m = Q/RE, V = m v1, P = m w, COP = RE/w.

Same answer, step by step with reasons

Step (write this)	Why / what it means
`Read` p_e `= 3.55 bar at -10 °C,` p_c `= 17.29 bar at 45 °C`	Saturation pressures from the R22 table.

h1 = 408.2 kJ/kg (vapour at 0 °C = -10 + 10 superheat), v1 = 0.0686 m^3/kg	Compressor suction state.
h2 = 450.4 kJ/kg (follow constant s from 1 to 17.29 bar)	Isentropic compression.
h3 = h4 = 249.6 kJ/kg (liquid at 40 °C = 45 - 5 subcool)	Throttling keeps h constant.
RE = h1 - h4 = 158.6 kJ/kg; w = h2 - h1 = 42.2 kJ/kg	Cooling per kg and work per kg.
m = 80/158.6 = 0.5045 kg/s	Total cooling / cooling per kg.
V = m v1 = 0.0346 m^3/s; COP = 158.6/42.2 = **3.75**; P = m w = **21.3 kW**	Volume flow, COP, power.

> **Memory trick: "h1, h2, h3 then m"**: RE = h1 − h3, w = h2 − h1, m = Q/RE.

A14 Thermostatic expansion valve Good to have Possible · 50% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(b)** [10] Explain the thermostatic expansion valve with a neat sketch.
- **2021 Q4(a)** [12] What is the function of expansion valve in a refrigeration system? Explain the working principle of thermostatic expansion valve.

Before you start: expansion valve

The expansion valve controls how much refrigerant enters the evaporator. A **thermostatic expansion valve (TEV)** senses the superheat at the evaporator outlet with a bulb: too much superheat (evaporator starving) → opens more; too little → closes, so liquid never reaches the compressor.

Not covered in the uploaded slides; written from standard textbooks (Arora, Khurmi). Check against class notes.

Function of an expansion valve

1. Reduce the pressure of liquid refrigerant from condenser pressure to evaporator pressure (throttling).
2. Control the flow of refrigerant into the evaporator to match the load.

Thermostatic expansion valve (TEV)

- Parts: valve body and needle, diaphragm, adjustable spring, **feeler bulb** clamped to the evaporator outlet, capillary tube joining bulb to the top of the diaphragm.
- Forces: bulb pressure p_b (top of diaphragm) opens the valve; evaporator pressure p_e + spring pressure p_s (below) close it. At balance $p_b = p_e + p_s$.
- It keeps a constant **degree of superheat** (about 5 K) at the evaporator outlet. When load rises, the vapour leaving gets more superheated, the bulb gets warmer, p_b rises, the valve opens and more refrigerant flows. When load falls the opposite happens.
- So the evaporator is kept well fed and no liquid reaches the compressor. Used in most DX systems.

Sketch: draw the valve with diaphragm at top, spring below, needle in the liquid line, capillary to a bulb on the suction line after the evaporator.

Remember: TEV (thermostatic expansion valve): bulb pressure opens; evaporator + spring close. Keeps superheat constant.

A16 Linde air liquefaction Skip if short Possible · 20% Medium

PAST-PAPER QUESTIONS

- **2021 Q3(c)** [10] Describe the Linde system of liquefaction of air.

Before you start: liquefying air

The Linde process: compress air, cool it, then expand it through a throttle valve; real gases cool on throttling (Joule-Thomson effect). A counter-flow heat exchanger pre-cools the incoming air with the cold returning gas, until part of it becomes liquid.

Not covered in the uploaded slides; written from standard textbooks (Arora, Khurmi). Check against class notes.

Linde system of air liquefaction (2021 3c)

1. Atmospheric air is compressed to about 200 bar in a multistage compressor with intercooling, then cooled to ambient in an after-cooler (nearly isothermal compression).
2. High-pressure air passes through a counter-flow heat exchanger where it is cooled by the cold unliquefied air returning from the separator.
3. It is then throttled through an expansion valve (Joule-Thomson effect); the temperature drops further and a fraction y of air liquefies.
4. Liquid is collected in a separator; the cold vapour returns through the heat exchanger (pre-cooling the incoming air) and is mixed with make-up air at the compressor inlet.

```
Yield y = (h1 - h2)/(h1 - hf)  (1 = low-pressure gas at ambient, 2 = high-pressure gas at ambient, f = saturated liquid)
```

Throttling cools only because air is below its inversion temperature. The process starts slowly and becomes continuous once the heat exchanger has cooled down.

Remember: Linde: compress, cool, regenerate in HX, throttle (J-T), separate liquid.

A11 Ammonia ice-plant numerical (COP % of theoretical) Good to have

Possible · 50% Hard

PAST-PAPER QUESTIONS

- **2023 Q2(c)** [12] An ammonia refrigerator produces 25 tonnes of ice from and at 0 °C in 24 hours. The temperature range of the compressor is from 15 °C to −5 °C. The vapor is dry saturated at the end of compression and an expansion valve is used. Assume a COP (coefficient of performance) of 45% of the theoretical value. Calculate the power required to drive the compressor. Latent heat of ice is 335 kJ/kg.
- **2021 Q2(c)** [15] Same as above with 30 tonnes of ice, compressor range 20 °C to −10 °C, COP = 55% of theoretical.

Before you start: ice plant numerical idea

Find the cooling load (kW) from the ice made per day. Find the theoretical COP from ammonia table values (dry saturated after compression means state 2 is on the vapour line, so s1 = s2 = s_g at condenser temperature). Actual COP = given % × theoretical. Power = load / actual COP.

Dryness fraction x = fraction of vapour in a wet mixture: $s = s_f + x(s_g - s_f)$, $h = h_f + x(h_g - h_f)$.

Method (from the slides' worked example, 30 t ice, 25 °C to −15 °C, COP 60%: answer 33.2 kW)

1. Load: 1 tonne of ice per day from and at 0 °C = 1000 × 335/86400 = 3.88 kW.
2. Dry saturated at the end of compression: $s_1 = s_2 = s_g$ at condenser temperature. Find x_1 from $s_1 = s_{f1} + x_1(s_{g1} - s_{f1})$, then $h_1 = h_{f1} + x_1(h_{g1} - h_{f1})$.
3. $COP_{th} = (h_1 - h_{f3})/(h_2 - h_1)$; COP_{actual} = % × COP_{th}; $P = Q/COP_{actual}$.

2023 2(c): 25 t ice/day, 15 °C to −5 °C, COP = 45% of theoretical

NH_3 table (same reference as the slides): at 15 °C h_f = 251.4, h_g = 1457.5 kJ/kg, s_g = 5.1493; at −5 °C h_f = 158.4, h_g = 1437.6 kJ/kg, s_f = 0.6313, s_g = 5.4019 kJ/kg K

$x_1 = (5.1493 - 0.6313)/(5.4019 - 0.6313) = 0.947$

$h_1 = 158.4 + 0.947(1437.6 - 158.4) = 1369.9$ kJ/kg

$COP_{th} = (1369.9 - 251.4)/(1457.5 - 1369.9) = 1118.5/87.6 = 12.77$

$COP_{actual} = 0.45 \times 12.77 = 5.75$

Q = 25 × 1000 × 335/86400 = 96.9 kW

Power = 96.9/5.75 = 16.9 kW

2021 2(c): 30 t ice/day, 20 °C to −10 °C, COP = 55% of theoretical

At 20 °C h_f = 275.0, h_g = 1461.3, s_g = 5.0910; at −10 °C h_f = 135.5, h_g = 1431.7, s_f = 0.5454, s_g = 5.4712

$x_1 = (5.0910 - 0.5454)/(5.4712 - 0.5454) = 0.923$; $h_1 = 135.5 + 0.923(1296.2) = 1331.7$ kJ/kg

$COP_{th} = (1331.7 - 275.0)/(1461.3 - 1331.7) = 1056.7/129.6 = 8.15$; actual = 0.55 × 8.15 = 4.48

Q = 30 × 3.877 = 116.3 kW

Power = 116.3/4.48 = 26.0 kW

Use the NH_3 table given in the exam; values above may differ by a few kJ/kg but the COP will be very close.

Remember: Ice tonnes × 3.88 kW. x_1 from entropy, h_1 from x_1, then COP × %.

Same answer, step by step with reasons

Step (write this)	Why / what it means
`Q = 25 t/day × 1000 × 335 / 86,400 = 96.9 kW`	Ice made per second × latent heat. 86,400 s per day.
`x1 = (`s_g `at 15 °C -` s_f `at -5 °C)/(`s_{fg} `at -5 °C) = 0.947`	Entropy same before and after compression; state 2 is dry saturated at 15 °C.
`h1 = 158.4 + 0.947 × 1279.2 = 1369.9 kJ/kg`	Enthalpy of the wet vapour entering the compressor.
`COP`$_{th}$ `= (1369.9 - 251.4)/(1457.5 - 1369.9) = 12.77`	(h1 − h_{f3}) / (h2 − h1).
`COP`$_{actual}$ `= 0.45 × 12.77 = 5.75; Power = 96.9/5.75 =` **`16.9 kW`**	Real machine reaches only 45 % of ideal.

Memory trick: 1 tonne ice/day ≈ **3.88 kW**.

Section B: Air conditioning

B1 Psychrometric definitions

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2024 Q5(a)** [12] Define: (i) Humid specific heat, (ii) Sensible heat factor, (iii) Degree of saturation, (iv) Bypass factor.
- **2023 Q5(a)** [12] Define: (i) Specific humidity, (ii) Degree of saturation, (iii) Wet bulb temperature, (iv) Humid specific heat.
- **2023 Q5(b)** [6] What is meant by SHF? How SHF line is drawn in psychrometric chart?
- **2021 Q5(a)** [12] Define: (i) Dew point temperature, (ii) Relative humidity, (iii) Wet bulb temperature, (iv) Sensible heat factor.
- **2020 Q4(a)** [10] What is meant by bypass factor? How do you draw sensible heat factor (SHF) line on a psychrometric chart?

Before you start: psychrometry in one minute

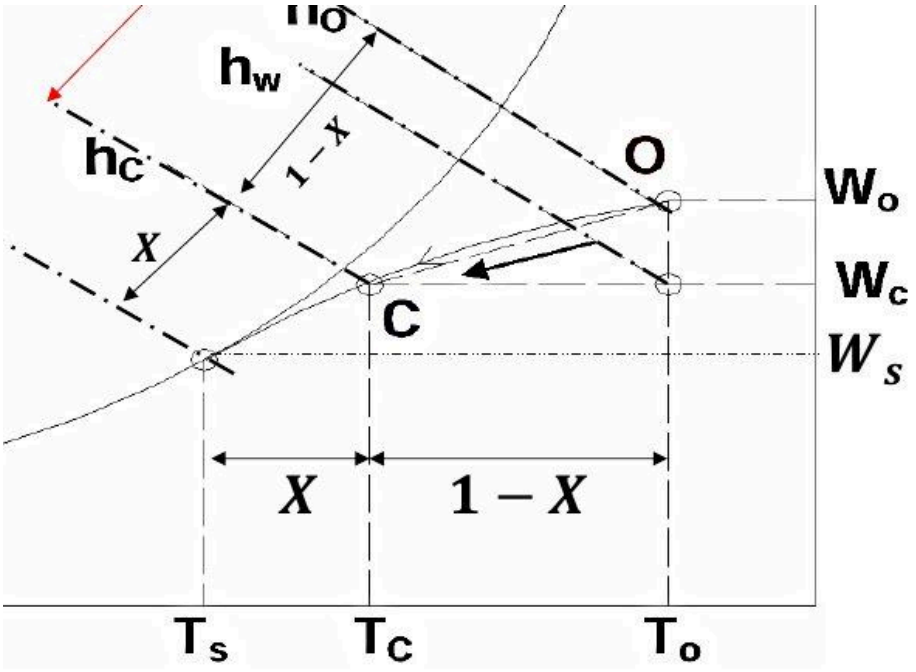
Psychrometry = study of moist air (dry air + water vapour). **Dry bulb temperature (DBT)** = normal thermometer reading. **Wet bulb (WBT)** = reading with a wet wick (lower, because water evaporates). **Dew point (DPT)** = temperature at which water starts to condense. **Specific humidity ω** = kg water per kg dry air. **Relative humidity (RH)** = how close the air is to saturation (%).

The **psychrometric chart** plots all of these; you find a point from any two values.

Each term carries 3 marks. Write the definition, the formula, and one line on why it matters.

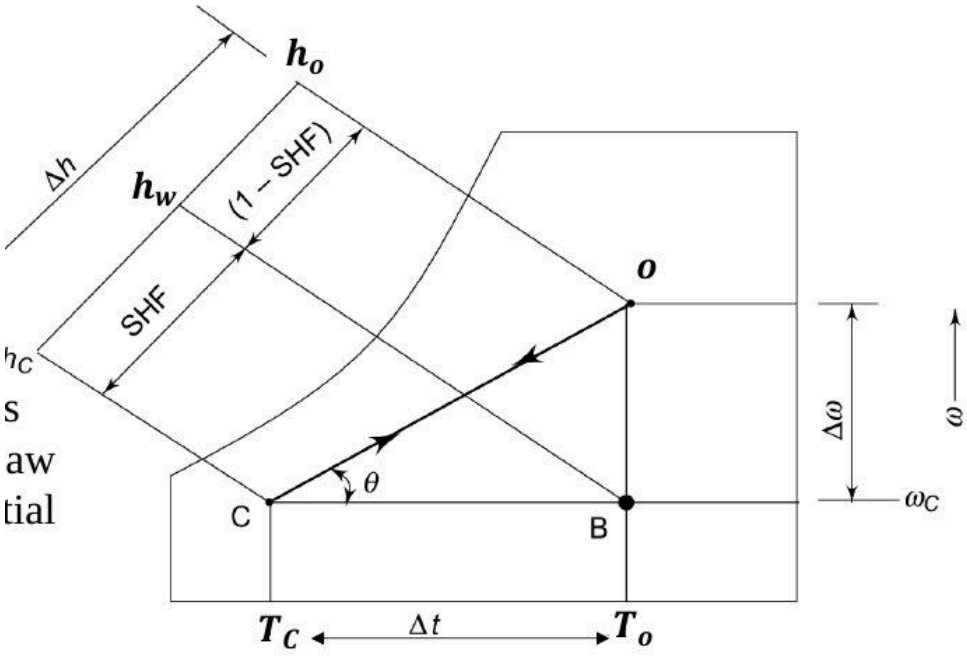
Term	Definition	Formula
Dry bulb temp (DBT)	Temperature of moist air read by an ordinary thermometer.	–
Wet bulb temp (WBT)	Lowest temperature reached by evaporating water into the air, read by a thermometer whose bulb has a wet wick in moving air. DBT (dry bulb temperature) − WBT (wet bulb temperature) = wet bulb depression; larger depression means drier air. Saturated air: WBT = DBT.	DPT (dew point temperature) ≤ WBT ≤ DBT
Dew point temp (DPT)	Temperature at which moisture starts to condense when unsaturated air is cooled at constant pressure (constant w). Saturated air: DPT = DBT.	$p_{sat}(DPT) = p_v$
Specific humidity / humidity ratio (w)	Mass of water vapour per kg of dry air. Does not change with temperature unless moisture is added or removed.	$w = 0.622\ p_v/(p_t - p_v)$ kg/kg d.a.
Relative humidity (φ)	Ratio of the vapour partial pressure to the saturation pressure at the same DBT; shows how close air is to saturation. Strongly depends on temperature.	$\varphi = p_v/p_{sat}$
Degree of saturation (μ)	Ratio of the humidity ratio to the humidity ratio of saturated air at the same temperature and pressure. 0 for dry air, 1 for saturated air; nearly equal to φ.	$\mu = w/w_s = \varphi(p_t - p_s)/(p_t - p_v)$
Humid specific heat (c_{pm})	Heat needed to raise 1 kg of dry air plus its water vapour by 1 K.	$c_{pm} = 1.005 + 1.88w \approx 1.0216$ kJ/kg K
Enthalpy (h)	Enthalpy of dry air + enthalpy of its vapour per kg dry air.	$h = 1.005t + w(2501 + 1.88t)$ kJ/kg d.a.
Sensible heat factor (SHF)	Ratio of sensible heat to total heat of a process. 1 = pure sensible, 0 = pure latent; 0.75-0.80 normal, ~0.6 in humid climates.	$SHF = Q_s/(Q_s + Q_L)$
Bypass factor (X)	Fraction of air passing the coil that does not touch it and leaves at inlet state; exit air is a mix, so it never reaches ADP (apparatus dew point). Contact factor = 1 − X. Lowered by more coil rows, lower air velocity, closer fins.	$X = (t_c - t_s)/(t_o - t_s)$
Apparatus dew point (ADP)	Effective coil surface temperature; where the condition (SHF) line meets the saturation curve.	$t_s = t_{ADP}$

Bypass factor on the chart



Bypass factor X and contact factor 1–X: air O is cooled toward ADP (s) but leaves at C (class 03 slides)

How to draw the SHF line on the psychrometric chart (2023 5b, 2020 4a)



SHF line O-C: slope fixed by SHF only (class 03 slides)

1. You need two things: the initial state (room point) and the SHF.
2. On the chart's protractor (SHF scale), join the reference point (26 °C, 50% RH (relative humidity)) to the given SHF value.
3. Draw a line parallel to it through the room state point. That line is the SHF line; extend it to meet the saturation curve at the ADP.

```
tan θ = Δw/ΔT = (1/2451)(1 - SHF)/SHF   (slope depends only on SHF)
```

Psychrometry is the study of the properties of moist air (mixture of dry air and water vapour) and the processes that change them.

Remember: Three temperatures: DBT ≥ WBT ≥ DPT (all equal when saturated). $w = 0.622\ p_v/(p_t - p_v)$. $SHF = Q_s/Q_t$. $X = (t_c - t_s)/(t_o - t_s)$.

B4 Thermal comfort and the ASHRAE comfort chart Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2024 Q6(a)** [5] Draw the ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) comfort chart.
- **2024 Q6(b)** [5] Is it impossible to satisfy a thermal environment that satisfies everyone? Explain with the equation: $PPD = 100 - 95 \times e^{-0.03353\ PMV^4 - 0.2179\ PMV^2}$
- **2023 Q6(a)** [5] Draw the ASHRAE comfort chart.
- **2023 Q6(b)** [6] What is meant by indoor design conditions and outdoor design conditions?
- **2021 Q6(a)** [8] What is meant by thermal comfort? What are the factors affecting thermal comfort?
- **2020 Q4(b)** [10] What is meant by thermal comfort? What are the factors affecting thermal comfort?
- **2020 Q4(c)** [6] Distinguish between effective temperature and operative temperature.
- **2020 Q4(d)** [4] Draw the ASHRAE comfort chart.

Before you start: comfort

People feel comfortable around 23-26 °C with 40-60 % RH (relative humidity) and gentle air movement. The **effective temperature** combines temperature, humidity and air speed into one number; the ASHRAE comfort chart shows the comfort zone on psychrometric axes.

Thermal comfort and its factors (2021 6a, 2020 4b)

Thermal comfort is the condition of mind which expresses satisfaction with the thermal environment (ASHRAE 55). The body is comfortable when its heat production equals its heat loss without strain.

1. **Environmental:** air (dry bulb) temperature, mean radiant temperature (MRT), relative humidity, air velocity.
2. **Personal:** activity level (metabolic rate, met) and clothing insulation (clo).
3. Others: age, sex, health, acclimatization, air purity.

ASHRAE comfort chart (5 marks, asked 3 times)

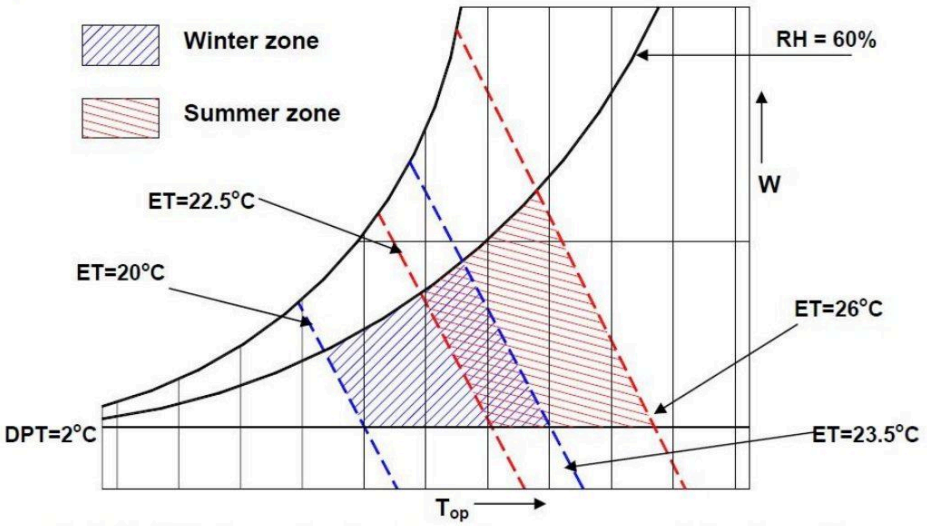


Fig.29.3: *ASHRAE comfort chart for a sedentary person (activity ≈ 1.2 met)*

ASHRAE comfort chart for a sedentary person, 1.2 met (course slides)

- Drawn on psychrometric axes: operative temperature on x, humidity ratio on y.
- Comfort zones are bounded by **effective temperature (ET) lines**, the **60% RH** line on top and **2 °C dew point** at the bottom.
- **Winter zone:** ET 20 to 23.5 °C. **Summer zone:** ET 22.5 to 26 °C. In the overlap, people in winter clothes feel slightly warm and in summer clothes slightly cool.
- Humidity limits avoid dry skin, eye irritation, respiratory problems and microbial growth.

Effective vs operative temperature (2020 4c)

Effective temperature (ET)	Operative temperature (t_o)
Temperature of a standard environment at 50% RH that gives the same heat loss from the skin (same feeling) as the actual environment	Uniform temperature of an imaginary black enclosure in which a person exchanges the same heat by radiation + convection as in the actual space
Combines air temp, humidity (and air motion)	Combines air temp and mean radiant temp; humidity not included
Lines of constant ET bound the comfort zones	$t_o \approx (t_a + t_{mrt})/2$ at low air speed; x-axis of comfort chart

PMV-PPD (2024 6b): can everyone be satisfied?

$$PPD = 100 - 95 \exp(-0.03353\ PMV^4 - 0.2179\ PMV^2)$$

PMV (predicted mean vote) is the average vote on the 7-point scale (−3 cold … 0 neutral … +3 hot). PPD is the % of people dissatisfied. Put the best case PMV = 0: PPD = 100 − 95 e^0 = **5%**. So even in the ideal environment at least 5% of people are still dissatisfied, because people differ in clothing, activity and preference. **It is impossible to satisfy everyone**; ASHRAE aims for PPD < 10% (−0.5 < PMV < +0.5).

Indoor and outdoor design conditions (2023 6b)

- **Indoor design conditions:** the DBT (dry bulb temperature), RH and air velocity to be maintained inside for comfort or process, chosen from the comfort chart. Typical summer comfort: 24-26 °C DBT, 50-60% RH, 0.15-0.25 m/s air speed; winter: 20-22 °C.
- **Outdoor design conditions:** the DBT and WBT (and daily range) of the location used for sizing equipment, taken from long-term weather data (e.g. 1%, 2.5% values that are exceeded only that fraction of hours). Not the absolute worst day, to avoid oversizing. Example: Dhaka summer ≈ 36 °C DBT / 28 °C WBT (wet bulb temperature).

Remember: Comfort chart: ET lines + 60% RH top + 2 °C DPT (dew point temperature) bottom; summer 22.5-26, winter 20-23.5. PMV 0 still gives PPD 5%.

B5 Indoor air quality (IAQ) and ventilation

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2024 Q6(c)** [15] Dhaka has twice the CO_2 concentration of Khulna (400 ppm). If a class room with 30 students is in a room of same size in both cities, determine the required ventilation rate for each location. Also, explain how to control IAQ (indoor air quality) in a classroom.
- **2023 Q6(d)** [12] What are effects of IAQ on health and comfort? How to control IAQ?
- **2021 Q6(d)** [10] Write short notes on: (i) Filtration, and (ii) Dilution.
- **2020 Q5(c)** [6] What are the effects of indoor air quality on health and comfort? How to control indoor air quality?

Before you start: fresh air

Indoor air gets stale (CO_2 from breathing, odours, VOCs). Ventilation brings in outside air to dilute it, typically 7.5-10 L/s per person, keeping CO_2 below about 1000 ppm.

IAQ and its effects on health and comfort (2023 6d, 2020 5c)

Indoor air quality (IAQ) is the condition of air inside buildings as it affects the health, comfort and well-being of occupants. People spend about 90% of time indoors, and indoor air can be 2-5 times more polluted than outdoor air (US EPA).

- **Sick building syndrome (SBS):** acute discomfort linked to time in a building, with no specific cause found; symptoms improve when the person leaves. Symptoms: cough, chest tightness, eye/nose/throat irritation, dry or itchy skin, headache, dizziness, nausea, fatigue, poor concentration.
- **Building-related illness (BRI):** medically identifiable disease that persists after leaving: allergic (asthma, rhinitis, hypersensitivity pneumonitis from mold, dust, VOCs), infections (Legionnaires' disease, TB, COVID-19), skin diseases (eczema, dermatitis).
- Pollutants: CO_2 and odours from people, VOCs from paints and furniture, particulates, mold spores, bacteria, CO/NO_x from combustion, radon.

How to control IAQ: three methods

1. **Source control** (most effective): low-VOC paints and furniture, no smoking, local exhaust; place outdoor intakes 8-10 ft above ground, away from loading docks and 10-15 ft from exhaust outlets; keep RH (relative humidity) below 60% to stop mold.
2. **Filtration:** remove particles 0.1-200 µm. Efficiency = $(C_{up} - C_{down})/C_{up} \times 100$. Filter types by MERV: panel (1-4), pleated (5-8), bag/pocket (9-15), HEPA (99.97% at 0.3 µm), ULPA (99.999% at 0.12 µm). Selection: cleanliness needed, dust type, pressure drop, space, cost, dust disposal.
3. **Dilution (ventilation):** bring in filtered outdoor air to dilute CO_2 and contaminants. By natural ventilation, forced (fan) ventilation, or infiltration (uncontrolled leakage). A CO_2 sensor can set the outdoor-air damper (demand-controlled ventilation).

Short notes (2021 6d): Filtration = removing particles and gases from supply air with filters (see list above); **Dilution** = lowering pollutant concentration by mixing in clean outdoor air (ventilation).

Numerical (2024 6c): classroom of 30 students, Khulna 400 ppm, Dhaka 800 ppm

CO_2 balance: $V \cdot C_{out} + N = V \cdot C_{space} \rightarrow$ **$V = n N_p / (C_{space} - C_{out})$**
Take N_p = 0.011 cfm CO_2 per person (1.2 met, seated students) and allowable indoor C_{space} = 1000 ppm (class 09 values).

Khulna: $V = 30 \times 0.011 / [(1000 - 400) \times 10^{-6}] = 0.33/0.0006 =$ **550 cfm** (18.3 cfm/person) = $550 \times 0.4719 =$ **260 L/s ≈ 0.26 m³/s**

Dhaka (C_{out} = 800 ppm): $V = 0.33 / [(1000 - 800) \times 10^{-6}] =$ **1650 cfm** (55 cfm/person) = **779 L/s ≈ 0.78 m³/s**
Dhaka needs 3 times the fresh air of Khulna, because the cleaner-air margin ($1000 - C_{out}$) is only 200 ppm instead of 600 ppm.

Controlling IAQ in the classroom: CO_2-sensor-controlled fresh-air damper or exhaust fan; filter the fresh air (MERV 8-13, since Dhaka's outdoor air is dusty); keep RH 40-60%; low-VOC materials; open windows in breaks for cross-ventilation; regular cleaning and filter replacement.

Remember: Source control, Filtration, Dilution. $V = nN_p/(C_{in} - C_{out})$.

B2 Psychrometric processes and the air washer Must study Very likely · 90% Easy

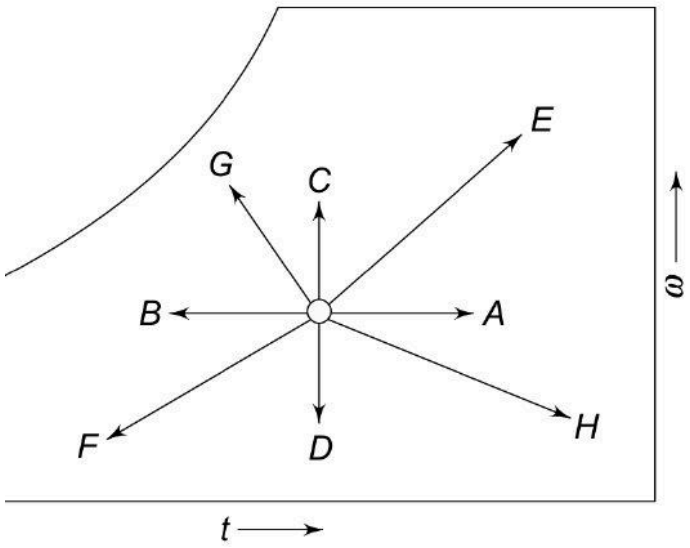
PAST-PAPER QUESTIONS

- **2024 Q5(b)** [8] What is an air washer? Describe the following process with an air washer: (i) Cooling and Dehumidification, (ii) Heating and Humidification.
- **2023 Q5(c)** [6] Explain the working principle of an air washer.
- **2021 Q5(b)** [8] What is psychrometry? Describe the following process in a psychrometric chart: (i) Cooling and dehumidification, (ii) Heating and humidification.

Before you start: processes on the chart

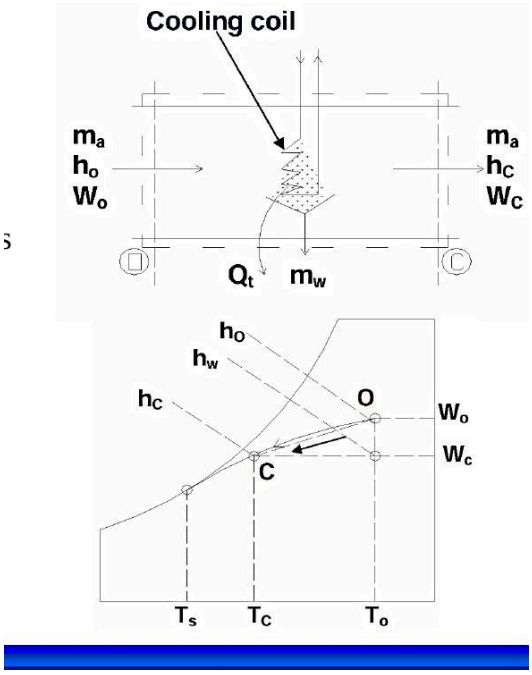
Heating/cooling only = horizontal line (ω constant). Cooling below dew point = dehumidification (moves left and down). Adding steam/water spray = humidification (moves up). Mixing two air streams = point on the straight line between them, closer to the bigger stream. An **air washer** sprays water into air.

Basic processes on the chart



Processes from state point: A sensible heating, B sensible cooling, C humidifying, D dehumidifying, E heating + humidifying, F cooling + dehumidifying, G cooling + humidifying, H heating + dehumidifying (class 03 slides)

(i) Cooling and dehumidification



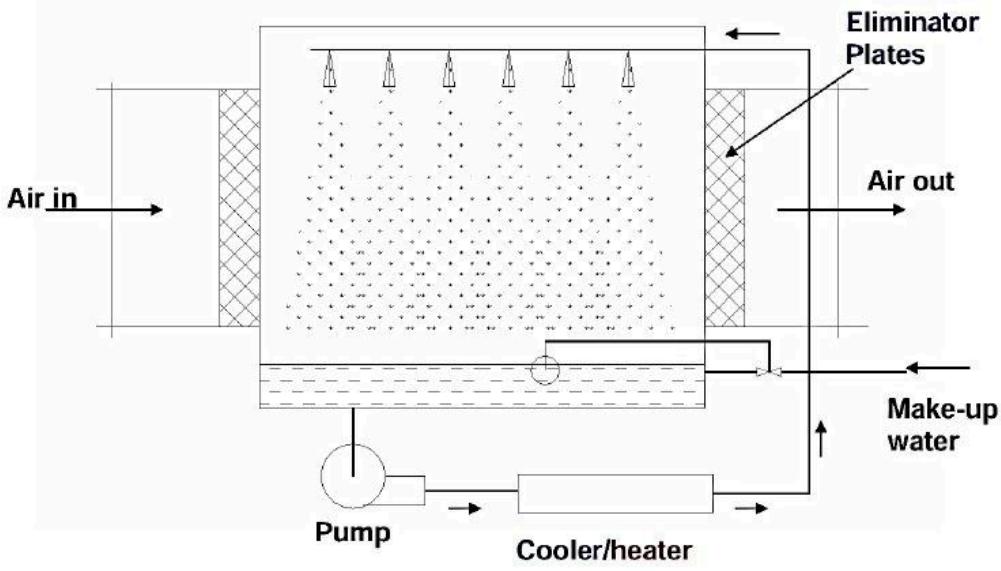
Cooling and dehumidification over a cooling coil, O to C (class 03 slides)

- Air is passed over a coil whose surface temperature is below the air's DPT (dew point temperature). Vapour condenses, so both DBT (dry bulb temperature) and w fall (process O→C, down and to the left).
- Water removed: $m_w = m_a(w_o - w_c)$. Coil load: $Q_t = m_a(h_o - h_c)$.
- $Q_s = m_a c_{pm}(t_o - t_c)$, $Q_L = m_a h_{fg}(w_o - w_c)$. Used in summer AC.

(ii) Heating and humidification

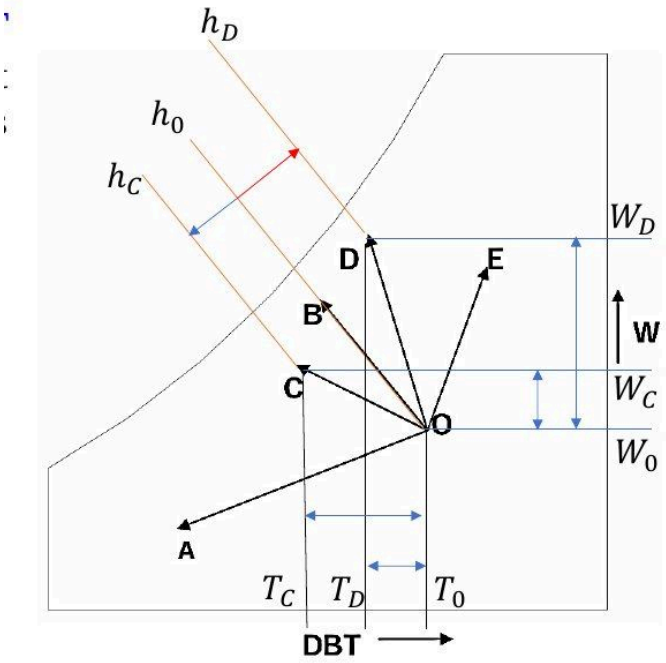
- Air is first sensibly heated by a heating coil, then water vapour (steam) or warm water spray is added. DBT and w both rise (up and to the right).
- $Q = m_a(h_2 - h_1) - m_w h_w$; used in winter AC for comfort.

Air washer (2024 5b, 2023 5c)



Air washer: spray nozzles, eliminator plates, pump and external cooler/heater (class 05 slides)

An **air washer** is a chamber in which air comes in direct contact with a spray of water; heat and moisture are exchanged between them. Eliminator plates stop water droplets leaving with the air; a pump recirculates the water through a cooler or heater; make-up water replaces evaporated water. The outlet air state is controlled by the **water temperature t_w**. Heat flows down the temperature gradient and moisture flows down the vapour pressure gradient.



Processes possible in an air washer, by water temperature (class 05 slides)

Water temperature	Process	Water must be
t_w < DPT	O-A Cooling and dehumidification	Externally cooled
t_w = DPT	Sensible cooling (w constant)	Cooled
DPT < t_w < WBT (wet bulb temperature)	O-C Cooling and humidification	Cooled
t_w = WBT	O-B Adiabatic (evaporative) cooling, h constant	Just recirculated
WBT < t_w < DBT	O-D Cooling and humidification, h rises	Externally heated
t_w > DBT	O-E Heating and humidification	Externally heated

With an air washer: cooling and dehumidification is done by spraying chilled water below the air's DPT; heating and humidification is done by spraying hot water above the air's DBT. Humidifying efficiency $\eta_H = (w_2 - w_1)/(w_s - w_1)$ = contact factor = 1 − X.

Remember: Air washer: the water temperature decides everything. Below DPT dries, at WBT is evaporative, above DBT heats.

B6 Air-conditioning systems: zone, all-air CAV / VAV, dual duct Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q7(a)** [8] What is meant by zone? Explain dual duct constant air volume air conditioning system with schematic diagram.
- **2024 Q6(d)** [10] Using the data given in the following table, make a system choice matrix and explain which system will be most appropriate for KUET academic building's thermal load.

Criteria	Cooling capacity	Temperature control	Zone timing	Initial cost	Operating maintenance	Energy consumption
Relative importance	08	09	10	05	09	08
Split-type performance	08	08	07	06	06	07
Rooftop unit performance	10	09	08	08	09	09

- **2023 Q6(c)** [12] Describe the various all-air central air-conditioning system.
- **2021 Q6(b)** [9] What is meant by zone? Explain the difference between constant air volume and variable air volume air-conditioning system with schematic diagram.
- **2021 Q6(c)** [8] What are the advantages and disadvantages of rooftop air-conditioning unit?
- **2020 Q5(a)** [9] Classify air-conditioning system. Explain the working principle of split air-conditioning system with schematic diagram.
- **2020 Q5(b)** [7] Distinguish between constant air volume and variable air volume of all-air central air-conditioning system. What are the advantages of all-air systems?

Before you start: AC system types

All-air systems send cooled air through ducts from a central air handling unit (AHU). **CAV (constant air volume)** (constant air volume) varies supply temperature; **VAV (variable air volume)** (variable air volume) keeps temperature fixed and varies the amount of air to each zone with dampers (saves fan energy). **Dual duct** = one hot and one cold duct mixed at each room.

The uploaded Section B slides do not cover air distribution systems, so this answer is from standard textbooks (Arora, McQuiston, ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers)). Check it against your class notes.

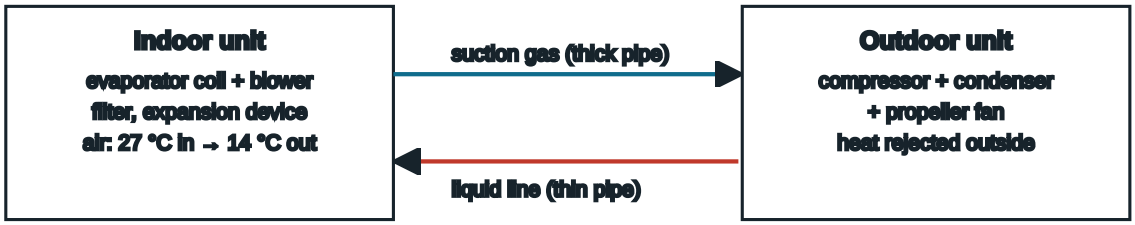
Zone

A **zone** is a space or group of spaces whose load pattern is similar enough to be controlled by **one thermostat** (one control point). Example: all west-facing offices form one zone; the interior core forms another.

Classification of AC systems (2020 5a)

- By purpose: comfort AC, industrial AC
- By season: summer, winter, year-round
- By equipment arrangement: unitary (window, split, packaged/rooftop) and central
- By medium to the room: all-air, all-water, air-water, refrigerant-based (DX, VRF (variable refrigerant flow))

Split AC working principle (2020 5a)



Split air conditioner, drawn for these notes.

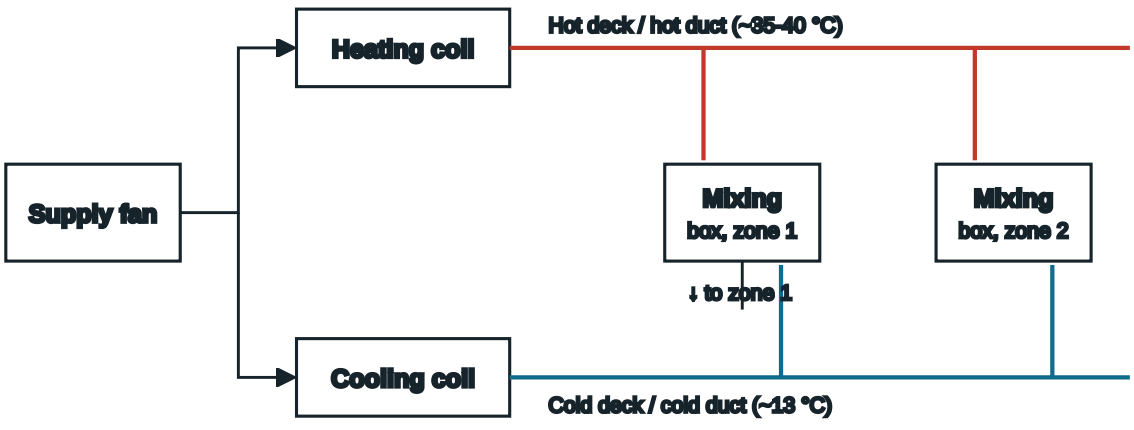
The **indoor unit** holds the evaporator coil, blower and filter; it cools and dehumidifies room air. The **outdoor unit** holds the compressor, condenser and fan, and rejects heat outside. Refrigerant flows between them in two copper pipes (liquid and suction). It follows the VCR (vapour compression refrigeration) cycle; quiet indoors because the compressor is outside; needs no ducts.

All-air central systems (2023 6c)

Air is the only medium that carries heating/cooling to the space; a central AHU (filter, coils, fan) conditions it and ducts deliver it.

1. **Single duct, constant air volume (CAV), single zone:** constant flow, supply temperature varied (or reheat) by one thermostat.
2. **Single duct CAV with terminal reheat:** air cooled to ~13 °C, reheated per zone; good control, wasteful.
3. **Multizone:** hot and cold decks at the AHU (air handling unit) mixed per zone by dampers; one duct per zone.
4. **Dual duct (CAV or VAV):** separate hot and cold ducts to mixing boxes at each zone.
5. **Variable air volume (VAV):** constant supply temperature, VAV boxes vary the airflow to each zone.

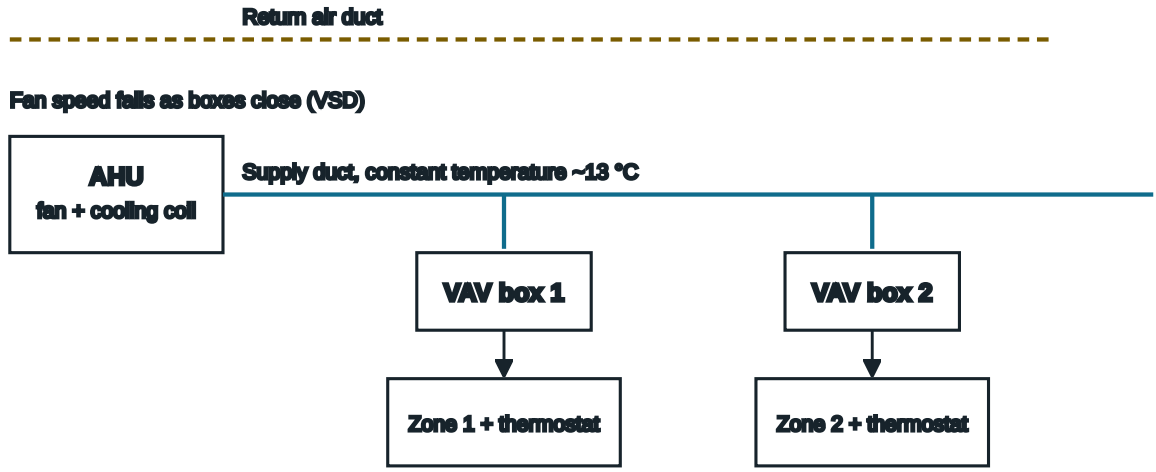
Dual duct CAV system (2024 7a)



Dual duct CAV system, drawn for these notes: one fan, a hot duct and a cold duct, and a mixing box per zone.

Return/fresh air is filtered and handled by one fan, then split into a **hot duct** (through a heating coil) and a **cold duct** (through a cooling coil). At each zone a **mixing box** with two linked dampers, worked by the zone thermostat, mixes hot and cold air in the right proportion. The total air to each zone stays constant; only its temperature changes. Advantages: each zone gets independent control at any time; heating and cooling available together. Disadvantages: two ducts take space and cost more; mixing hot and cold air wastes energy.

CAV vs VAV (2021 6b, 2020 5b)



VAV system, drawn for these notes: supply temperature is fixed and each zone's damper box varies the airflow.

Constant air volume (CAV)	Variable air volume (VAV)
Airflow constant; supply **temperature** varies with load	Supply temperature constant; **airflow** varies with load
Simple, low first cost, steady ventilation	VAV box per zone, variable-speed fan, more controls
Fan always at full power; reheat wastes energy	Fan power drops at part load; large energy saving
Suits single zone or uniform load	Suits multi-zone buildings with varying loads
Good air distribution at all loads	At low flow, ventilation and air distribution can suffer (minimum box setting needed)

Advantages of all-air systems: central equipment away from occupied space (easy maintenance, no water or drains in rooms); good humidity control and filtration; can use free cooling with outdoor air (economizer); simultaneous heating and cooling in different zones; quiet rooms. **Disadvantages:** large ducts and shafts, more fan energy, needs ceiling space.

Rooftop unit: advantages and disadvantages (2021 6c)

Advantages	Disadvantages
Self-contained, factory assembled and tested; quick to install	Roof must carry the weight; roof penetration can leak
Saves indoor floor space and plant room	Exposed to weather; shorter life
Noise and maintenance kept outside occupied area	Service work on the roof is awkward
Easy to add economizer, lower first cost	Lower efficiency than central chillers for large buildings; limited duct runs

System choice matrix (2024 6d)

Criteria	Cooling capacity	Temperature control	Zone timing	Initial cost	Operating maintenance	Energy consumption
Relative importance	08	09	10	05	09	08
Split-type performance	08	08	07	06	06	07
Rooftop unit performance	10	09	08	08	09	09

Data given in 2024 6(d)

Score = Σ (relative importance × performance)

Split: 8×8 + 9×8 + 10×7 + 5×6 + 9×6 + 8×7 = 64 + 72 + 70 + 30 + 54 + 56 = **346**

Rooftop: 8×10 + 9×9 + 10×8 + 5×8 + 9×9 + 8×9 = 80 + 81 + 80 + 40 + 81 + 72 = **434**

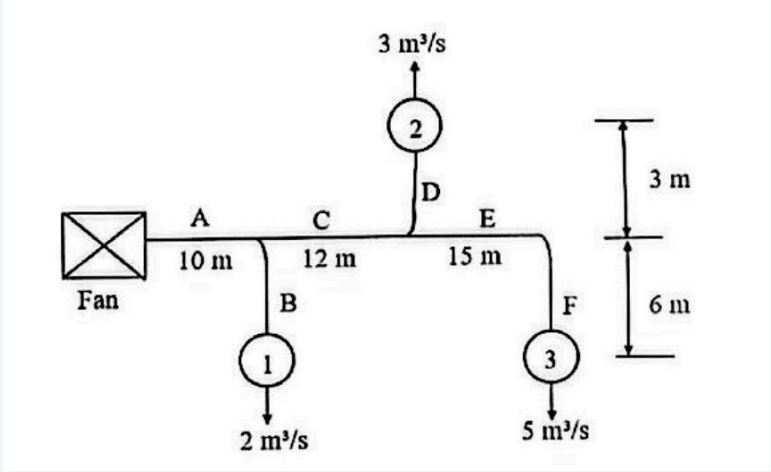
Rooftop unit scores higher (434 vs 346, out of 490) and is the better choice for the KUET academic building. It wins on cooling capacity, maintenance and energy, which matter for a large building with long operating hours.

Remember: CAV: fixed air, vary temperature. VAV: fixed temperature, vary air. Dual duct: hot + cold duct mixed per zone.

B9 Duct sizing numerical (equal friction method) Must study Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q8(c)** [15] For the duct system as shown in figure, determine the dimension of rectangular duct with aspect ratio 3:1. Assume all dynamic loss coefficients are 0.8.



- **2023 Q8(d)** [15] Determine the size of the duct system shown in figure using equal friction method assuming pressure drop of 0.1 mm H_2O/m length.

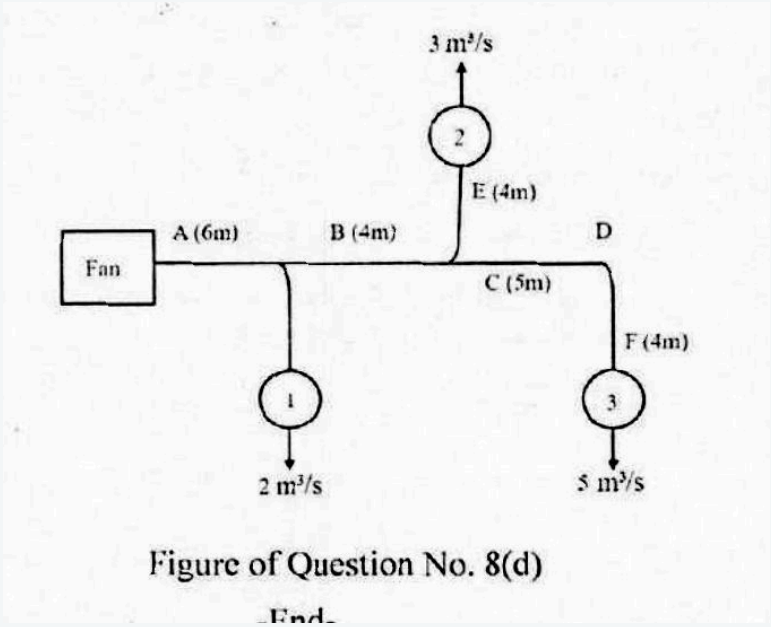
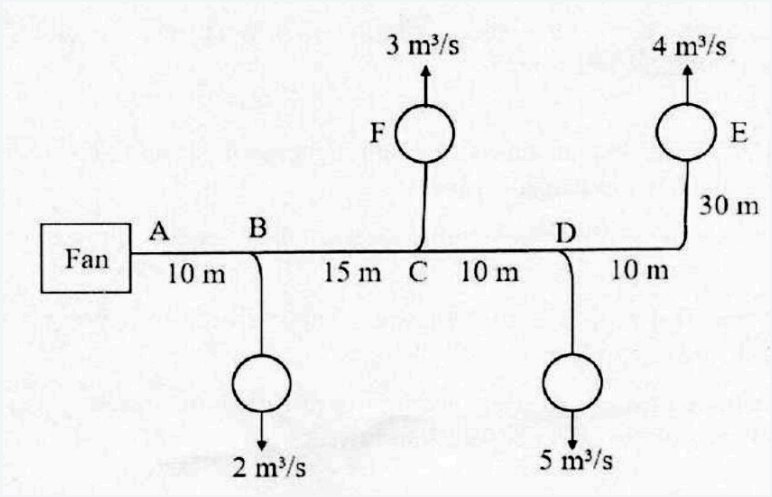


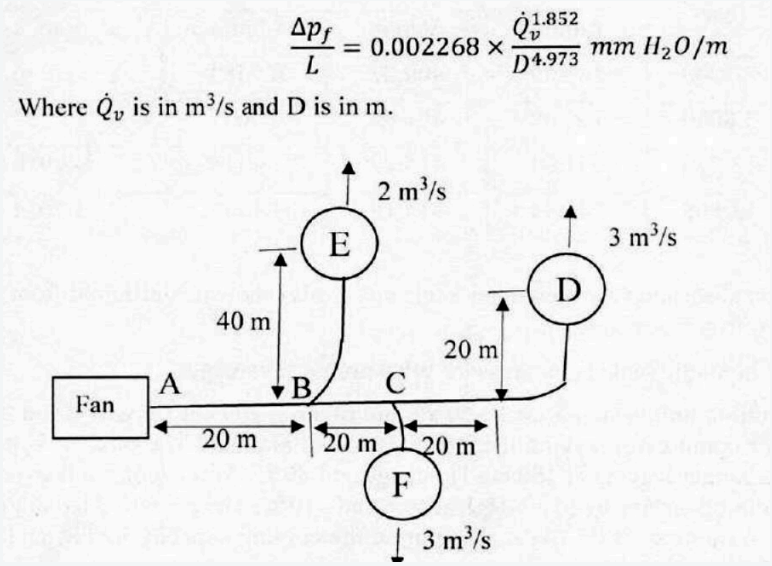
Figure of Question No. 8(d)

End

- **2021 Q8(d)** [14] For the duct system as shown in figure, determine the dimensions AB, BC, CD and DE of the rectangular ducts with aspect ratio 3:1 assuming friction rate of 0.1 mm H_2O/m length duct.



- **2020 Q6(c)** [12] Determine the dimensions of AB, BC, CD as shown in figure using equal friction method. Choose a friction rate of 0.09 mm H_2O/m. Use $\Delta p_f/L = 0.002268 \times Q^{1.852}/D^{4.973}$ mm H_2O/m (Q in m^3/s, D in m).



Before you start: duct sizing

Air needs a pressure push to flow through ducts; friction loses some pressure per metre. In the **equal friction method** every duct section is sized for the same friction loss per metre. Bigger flow → bigger diameter. For a rectangular duct use the equivalent diameter $D_e = 1.3(ab)^{0.625}/(a + b)^{0.25}$.

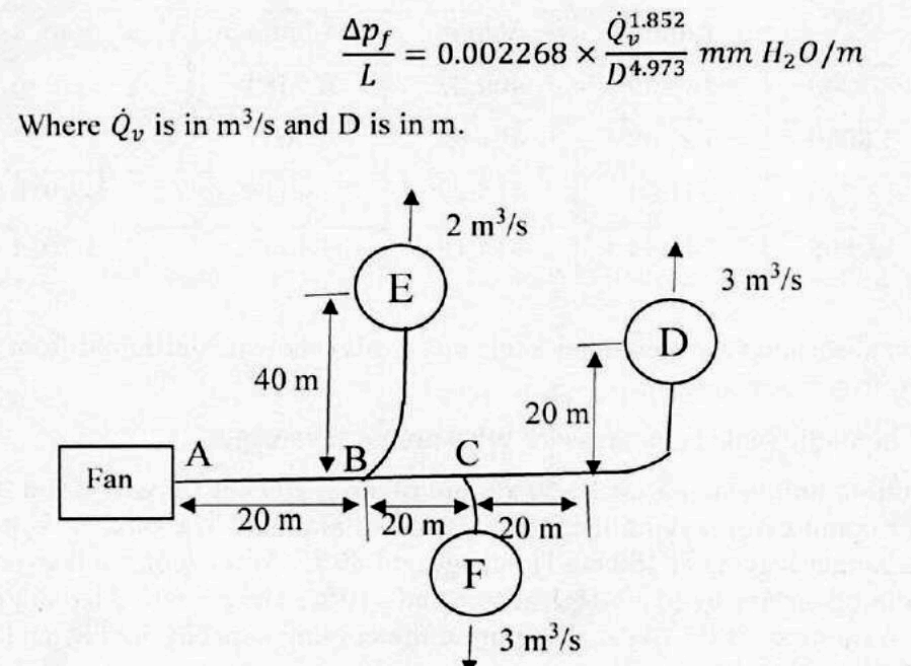
Duct design is not in the uploaded slides. The method below is the standard equal friction method (Arora ch. 22); the friction formula is the one printed on the 2020 paper. Rectangular sizes use the equivalent diameter $D_e = 1.3(ab)^{0.625}/(a + b)^{0.25}$.

Equal friction method in 5 steps

1. Write the airflow in each section by adding the outlet flows downstream of it.
2. Choose a constant friction rate f (given: 0.09 or 0.1 mm H_2O per m) for every section.
3. Find the round diameter from the friction chart or formula: $f = 0.002268\ Q^{1.852}/D^{4.973} \Rightarrow D = (0.002268\ Q^{1.852}/f)^{1/4.973}$

4. For a rectangular duct with a = 3b, solve $D_e = 1.3(3b^2)^{0.625}/(4b)^{0.25}$ for b. Quick rule: for 3:1, **b ≈ 0.548 D, a ≈ 1.643 D**.
5. Fan total pressure = f × (length of the longest run) + dynamic losses (K × velocity pressure) on that run.

2020 6(c): f = 0.09 mm H_2O/m (round ducts)



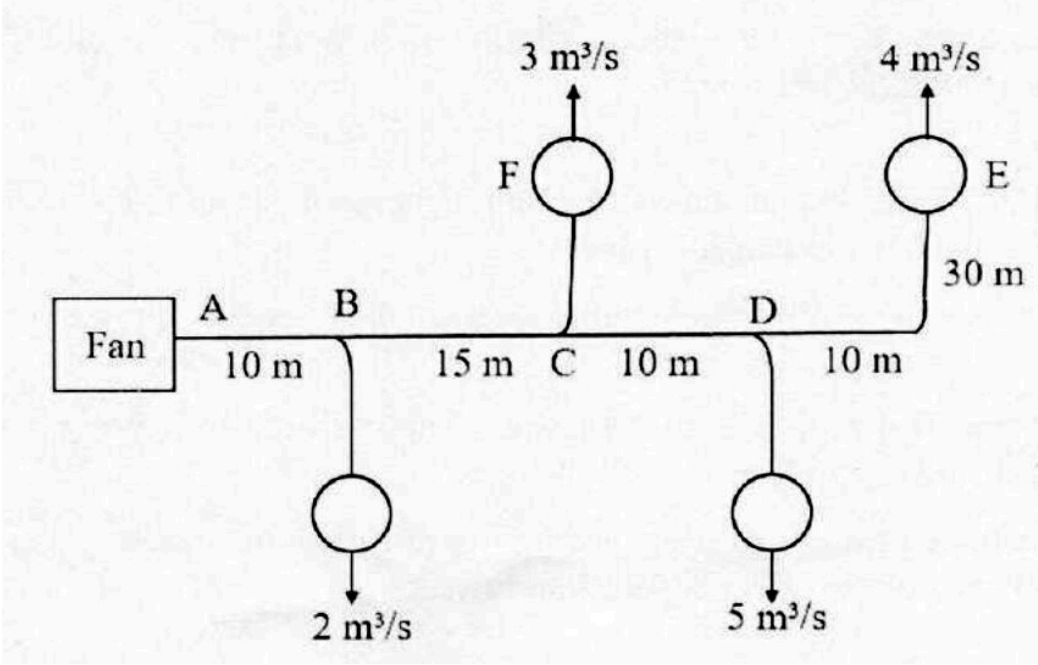
Duct layout, 2020

Flows: E = 2, F = 3, D = 3 m³/s. AB = 8, BC = 6, CD = 3 m³/s.

Section	Q (m³/s)	D (m)	V (m/s)
AB	8	**1.035**	9.5
BC	6	**0.930**	8.8
CD	3	**0.718**	7.4
Branch to E	2	0.618	6.7
Branch to F	3	0.718	7.4

Sample: $D_{AB} = (0.002268 \times 8^{1.852}/0.09)^{1/4.973} = (0.002268 \times 47.0/0.09)^{0.2011} = (1.184)^{0.2011} =$ 1.035 m

2021 8(d): f = 0.1 mm H_2O/m, rectangular 3:1



Duct layout, 2021

Outlets: B 2, F 3, D 5, E 4 m³/s. AB = 14, BC = 12, CD = 9, DE = 4 m³/s.

Section	Q	D_e (m)	a × b (m)
AB	14	1.248	**2.05 × 0.68**
BC	12	1.178	**1.94 × 0.65**
CD	9	1.059	**1.74 × 0.58**
DE	4	0.783	**1.29 × 0.43**

Index run A-B-C-D-E = 10 + 15 + 10 + 10 + 30 = 75 m → friction loss = 75 × 0.1 = 7.5 mm H_2O

2023 8(d): f = 0.1 mm H_2O/m

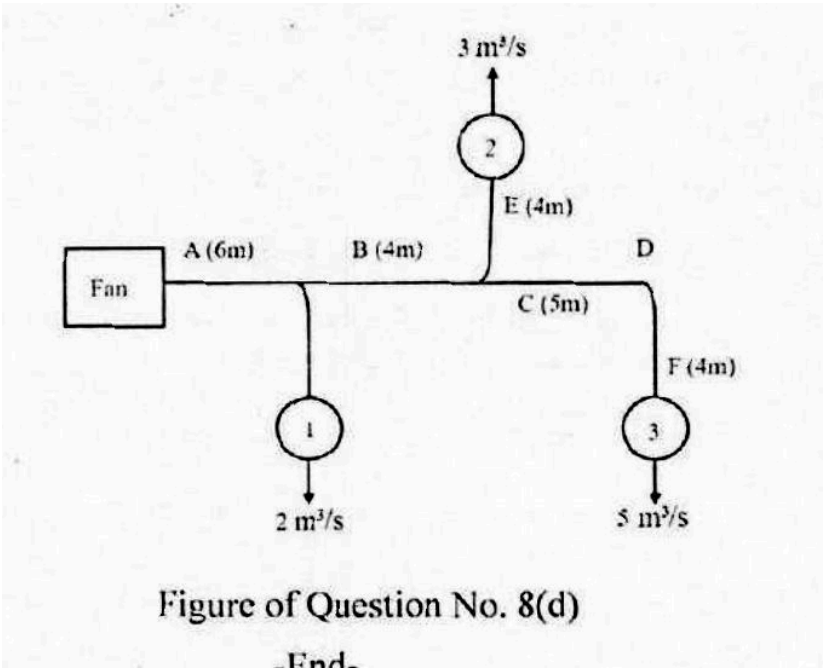


Figure of Question No. 8(d)

-End-

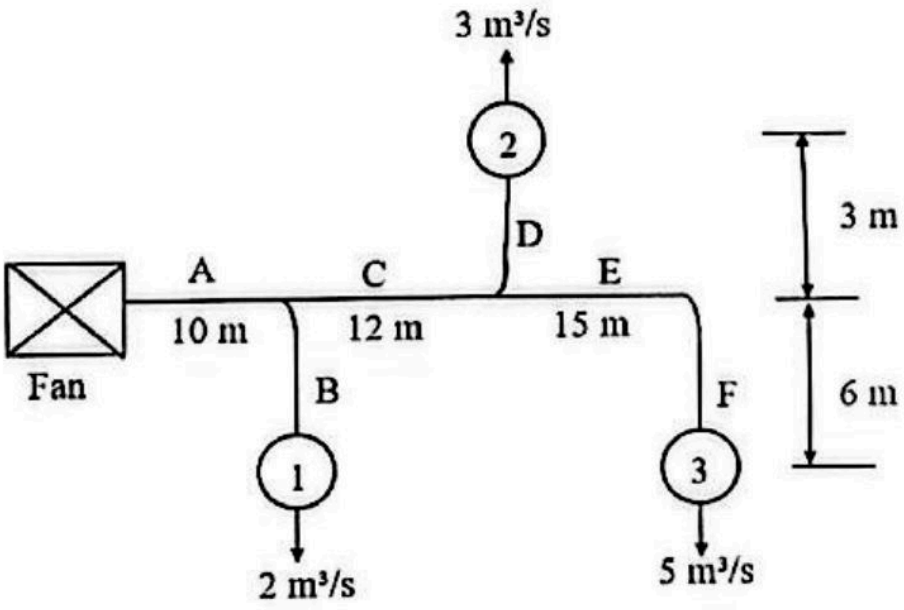
Duct layout, 2023

Outlets: 1 = 2, 2 = 3, 3 = 5 m³/s. A = 10, B = 8, C = 5, E = 3, F = 5 m³/s.

Section	Q	Round D (m)	Rect 3:1 (m)
A (6 m)	10	**1.10**	1.81 × 0.60
B (4 m)	8	**1.01**	1.66 × 0.55
C (5 m)	5	**0.85**	1.40 × 0.47
F (4 m)	5	**0.85**	1.40 × 0.47
E (4 m)	3	**0.70**	1.16 × 0.39
Branch to 1	2	**0.61**	0.99 × 0.33

Longest run A-B-C-F = 19 m → 1.9 mm H_2O friction.

2024 8(c): rectangular 3:1, dynamic loss coefficient 0.8



Duct layout, 2024

Friction rate is not given; assume **0.1 mm H_2O/m** (equal friction). Outlets: 1 = 2, 2 = 3, 3 = 5 m^3/s. A = 10, C = 8, E = 5, F = 5, B = 2, D = 3 m^3/s.

Section	Q	D_e (m)	a × b (m)	V (m/s)
A (10 m)	10	1.101	**1.81 × 0.60**	9.2
C (12 m)	8	1.013	**1.66 × 0.55**	8.7
E (15 m)	5	0.850	**1.40 × 0.47**	7.7
F (6 m)	5	0.850	**1.40 × 0.47**	7.7
B (6 m)	2	0.605	**0.99 × 0.33**	6.1
D (3 m)	3	0.703	**1.16 × 0.39**	6.8

Index run A-C-E-F = 43 m: friction = 4.3 mm H_2O

Dynamic losses, K = 0.8 at the two junctions and the elbow on this run: velocity pressure p_v = $\rho V^2/2$ (in mm H_2O = $1.2V^2/(2 \times 9.81)$): C 4.6, E 3.6, F 3.6 mm

0.8 × (4.6 + 3.6 + 3.6) = 9.4 mm H_2O

Fan pressure ≈ 4.3 + 9.4 = 13.7 mm H_2O

Remember: Add flows backward from the outlets → same f everywhere → D = $(0.002268Q^{1.852}/f)^{0.201}$ → 3:1 rectangle: b = 0.548D, a = 3b.

Same answer, step by step with reasons

Step (write this)	Why / what it means
`Add outlet flows from the far end back to the fan to get Q in each section`	Each duct carries all the air for the outlets beyond it.
`Choose friction rate f (e.g. 0.09 or 0.1 mm water per m), same for all sections`	Equal friction method.
$D = (0.002268\ Q^{1.852}/f)^{1/4.973}$	From the friction formula printed on the paper; Q in m^3/s, D in m.
$(0.002268 \times 8^{1.852}/0.09)^{0.2011} = 1.035$ m	Calculator: $8^{1.852}$ = 47.0.
`Rectangular 3:1: b ≈ 0.548 D, a ≈ 1.643 D`	Gives same friction as the round duct (equivalent diameter).
`Fan pressure = f × longest run length (+ fitting losses)`	The index (longest) run decides the fan.

Memory trick: "Add flows, fix friction, find D".

B7 VRF systems (two-pipe / three-pipe heat recovery) Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2024 Q7(b)** [12] What is meant by VRF (variable refrigerant flow) system? Explain three pipe heat recovery VRF systems with necessary sketches.
- **2023 Q7(b)** [7] Distinguish between multi-split system and VRF system.
- **2021 Q7(a)** [12] What is meant by VRF system? Explain three-pipe heat recovery VRF systems with required sketches.
- **2021 Q7(b)** [8] What are the differences between a heat pump multisplit system and a VRF multisplit system?
- **2020 Q5(d)** [8] What are the advantages of VRF system? Draw the block diagram of two-pipe heat recovery VRF system.

Before you start: VRF

Variable refrigerant flow: one outdoor unit with a variable-speed compressor feeds many indoor units by refrigerant pipes (no ducts, no chilled water). Two-pipe = all units cool or all heat; three-pipe heat recovery = some rooms can cool while others heat, moving heat between them.

VRF is not in the uploaded slides; this answer is from standard textbooks and manufacturer literature.

What is VRF

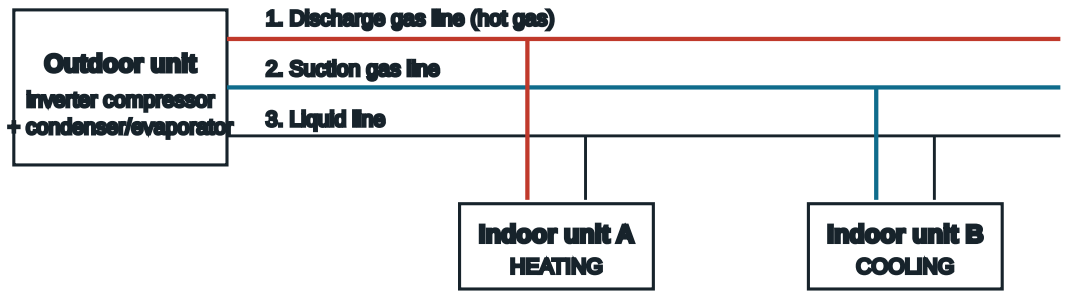
Variable refrigerant flow (VRF) is a ductless multi-split DX system in which one outdoor unit with an **inverter (variable speed) compressor** serves many indoor units through refrigerant piping. Electronic expansion valves at each indoor unit vary the refrigerant flow to match each zone's load.

Advantages (2020 5d)

1. High part-load efficiency: inverter compressor runs at exactly the needed capacity.
2. Individual zone control; up to 40-60 indoor units on one outdoor unit.
3. Heat recovery: simultaneous heating and cooling, rejected heat is reused.
4. No ducts or chilled-water pipes; small refrigerant pipes save space; modular, easy to expand.
5. Quiet indoor units; simple installation in existing buildings.
6. Individual energy metering possible.

Limitations: high first cost, large refrigerant charge (leak risk in small rooms), limited pipe lengths, fresh air needs a separate system.

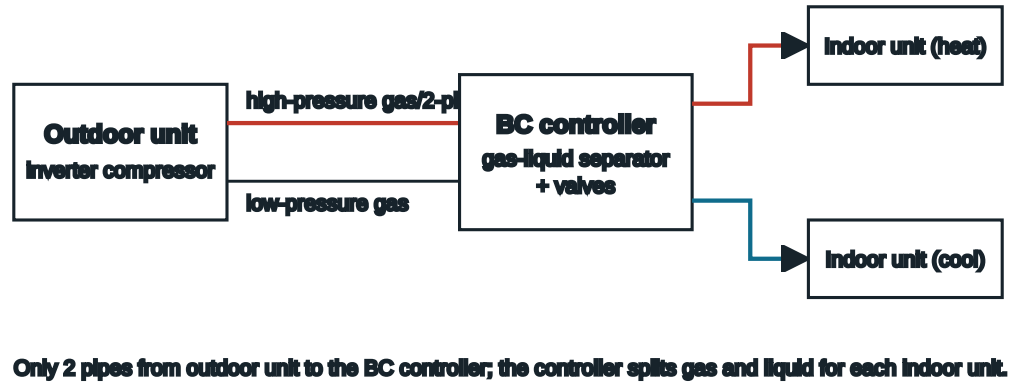
Three-pipe heat recovery VRF (2024 7b, 2021 7a)



Three-pipe heat recovery VRF, drawn for these notes. Solenoid valve boxes at each indoor unit pick the pipes.

- Three pipes run from the outdoor unit: **discharge (hot) gas line**, **suction gas line**, and **liquid line**.
- An indoor unit in **heating** takes hot gas from the discharge line, condenses it, and returns liquid to the liquid line.
- An indoor unit in **cooling** takes liquid from the liquid line, evaporates it, and returns gas to the suction line.
- So the heat removed from rooms being cooled is delivered to rooms being heated. The outdoor coil only handles the imbalance. Each indoor unit has a branch selector (solenoid valve) box.
- Result: COP (coefficient of performance) of heat recovery operation can reach 6-8 when loads balance.

Two-pipe heat recovery VRF (2020 5d)



Two-pipe heat recovery VRF block diagram, drawn for these notes.

Only two pipes (high-pressure two-phase line and low-pressure gas line) go from the outdoor unit to a **BC (branch) controller**. Its gas-liquid separator sends high-pressure gas to heating units and liquid to cooling units, then collects their returns. Fewer pipes and joints than three-pipe.

Multi-split vs VRF (2023 7b, 2021 7b)

Heat pump multi-split	VRF multi-split
Usually fixed-speed or simple inverter compressor	Inverter compressor(s), capacity modulates 10-100%
2-5 indoor units; each often has its own pipe pair to the outdoor unit	Up to ~60 indoor units on one pipe network with branches
All units in the same mode (all cool or all heat)	Heat recovery type can heat and cool at the same time
Short pipe lengths (~15-30 m)	Long pipes (up to ~150-1000 m total) and large height differences
On/off control, lower part-load efficiency	Electronic expansion valves, precise zone control, higher seasonal efficiency
Cheap, for homes and small offices	Costly, for commercial buildings

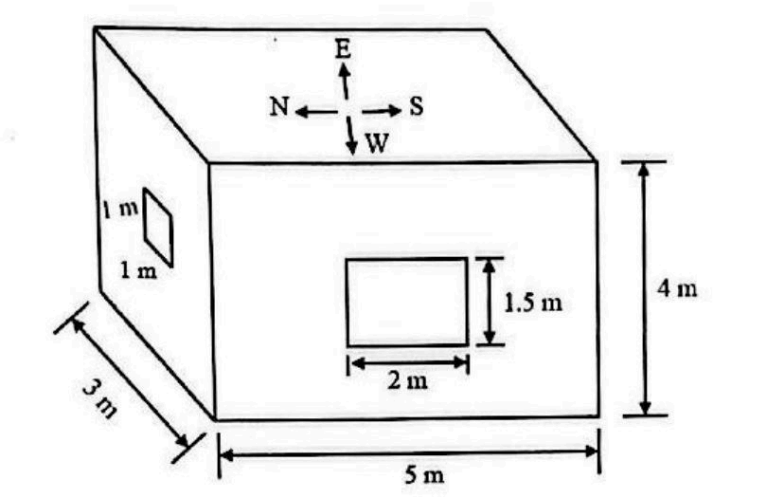
Remember: VRF = inverter + EEV per unit. Three pipes: hot gas, suction, liquid. Heating unit makes liquid, cooling unit uses it.

B8 Cooling load: components, assumptions, numerical Must study

Very likely · 100% Hard

PAST-PAPER QUESTIONS

- **2024 Q7(c)** [15] Two of the wall faces N and W contain double glaze window of a room as shown in figure. If there is no heat gain through other walls, what is the required cooling capacity of the room? Inside: DBT (dry bulb temperature) 43 °C, RH (relative humidity) 55% (as printed). Outside: DBT 43 °C, WBT (wet bulb temperature) 24 °C. U-value of wall, roof and floor 1.78, 1.3, 1.2. ETD (equivalent temperature difference) wall 25 °C, roof 30 °C. SHG 300 W/m², SC (shading coefficient) 0.86. Light 33 W/m² floor. Occupancy 4 (90 W sensible + 40 W latent each). Appliances 600 W sensible + 400 W latent. Infiltration 0.5 ACH (air changes per hour).



- **2023 Q7(c)** [7] What are assumptions should be considered while calculating the cooling load?
- **2023 Q7(d)** [15] Define heating and cooling load. Describe how the cooling load in the form of grand total load on air conditioning system can be evaluated.
- **2021 Q7(c)** [7] What is meant by cooling load? What assumptions behind cooling load design?
- **2020 Q6(a)** [10] What is cooling load? List the main components of cooling load estimation.

Before you start: cooling load

Cooling load = heat that must be removed per second to hold the room at set conditions. Sources: through walls/roof (sun + hot air), windows (sunshine through glass), outside air leaking in or ventilated, and inside: people, lights, equipment. **Sensible** heat changes temperature; **latent** heat is moisture.

Cooling load and heating load (2023 7d, 2021 7c, 2020 6a)

- **Cooling load:** the rate at which heat must be removed from a space to keep it at the design indoor condition. Peak occurs in the daytime; outdoor conditions and solar radiation change through the day, so it is an **unsteady** calculation; all internal gains add to it.
- **Heating load:** the rate of heat loss from the building in winter that must be supplied. Peak occurs before sunrise; usually calculated as **steady state**, with no solar gain, neglecting internal gains (conservative).

Assumptions in cooling load design (2023 7c, 2021 7c)

1. Design outside conditions are taken from long-term statistical weather data for the location, not any one actual year.

2. Solar load is estimated for **clear sky** conditions.
3. Building occupancy is at **full design capacity**.
4. All equipment and appliances run at a representative capacity.
5. Design load is near the maximum, not the absolute maximum (to avoid oversizing).

Components of cooling load (2020 6a)

External loads	Formula	Type
Walls, roof, floor, doors (opaque)	Q = U A CLTD (or ETD)	Sensible
Windows (fenestration)	$Q = A[U_g(T_o - T_i) + SHGF_{max} \cdot SC \cdot CLF]$ (solar heat gain factor; cooling load factor)	Sensible
Infiltration	$Q_s = m\, c_{pm}(T_o - T_i)$, $Q_L = m\, h_{fg}(W_o - W_i)$	Both
Bypassed ventilation air, duct gains	Q = m BPF (bypass factor) $c_{pm}\Delta T$, m BPF $h_{fg}\Delta W$	Both
Internal loads		
Occupants	N × sensible/person × CLF; N × latent/person	Both
Lighting	Watts × utilization factor × ballast factor (1.25 fluorescent) × CLF	Sensible
Equipment, appliances	Watts × utilization × CLF; latent part	Both

CLTD (cooling load temperature difference)/ETD: equivalent temperature difference that gives the same heat gain as the real combined effect of outdoor temperature, solar radiation and wall heat storage. Depends on orientation, construction, colour, outdoor temp, time, latitude. **SC** = solar gain through the glass / solar gain through standard clear glass. **CLF** accounts for the time lag of radiant gains.

Grand total load (2023 7d)

```
RSH (room sensible heat) = Σ sensible room gains; RLH (room latent heat) = Σ latent room gains; RTH (room total heat) = RSH + RLH

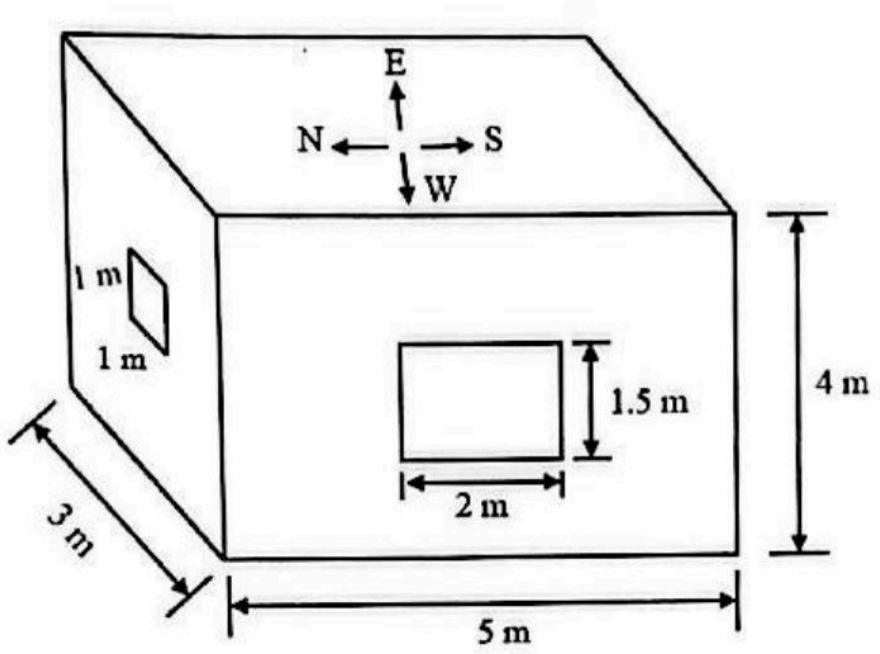
OATH (outside air total heat) = m_o(h_o - h_i) (ventilation load)

GTH (grand total heat) = RTH + OATH (+ safety factor and duct/fan gains, often ×1.1-1.25)

Plant capacity (TR) = GTH / 3.5 kW
```

Methods: rules of thumb (TR per m²), semi-empirical CLTD/CLF/SHGF method (hand calculation, used in exams), transfer function method and heat balance method (computer).

Numerical (2024 7c): room 5 m × 3 m × 4 m, windows on W and N walls



Room given in 2024 7(c)

Assumptions (same as the class 14 example the question is built from): inside 25 °C DBT, 50% RH (the printed "43 °C, 55%" must be a misprint); outside 43 °C DBT, 24 °C WBT; floor over a ventilated basement, so $\Delta T = 43 - 25 = 18$ K; glass U = 3.12 W/m²K (not given; class example value); CLF = 1; $\rho = 1.095$ kg/m³; $W_i = 0.00992$, $W_o = 0.0107$ kg/kg.

Areas (from the sketch: 5 m face is West with 2 × 1.5 m window; 3 m face is North with 1 × 1 m window):

West wall net = 5 × 4 − 3 = 17 m²; North wall net = 3 × 4 − 1 = 11 m²; roof = floor = 5 × 3 = 15 m²; glass = 4 m²; volume = 60 m³

1. Walls: 1.78 × (17 + 11) × 25 = **1246 W** (S)
2. Roof: 1.3 × 15 × 30 = **585 W** (S)
3. Floor: 1.2 × 15 × 18 = **324 W** (S)
4. Glass: 4 × [3.12 × 18 + 300 × 0.86] = 4 × 314.2 = **1257 W** (S)
5. Infiltration: $m = 1.095 \times 0.5 \times 60/3600 = 0.00913$ kg/s; $Q_S = 0.00913 \times 1021.6 \times 18$ = **168 W**; $Q_L = 0.00913 \times 2501 \times 10^3 \times (0.0107 - 0.00992)$ = **18 W**
6. People: 4 × 90 = **360 W** (S); 4 × 40 = **160 W** (L)
7. Lights: 33 × 15 = **495 W** (S)
8. Appliances: **600 W** (S), **400 W** (L)

RSH = 1246 + 585 + 324 + 1257 + 168 + 360 + 495 + 600 = **5035 W**; RLH = 18 + 160 + 400 = **578 W**

RTH = 5613 W; RSHF (room sensible heat factor) = 5035/5613 = 0.90

Required capacity with safety factor 1.25 = 5613 × 1.25 = 7016 W ≈ 2.0 TR (ton of refrigeration (3.5 kW))

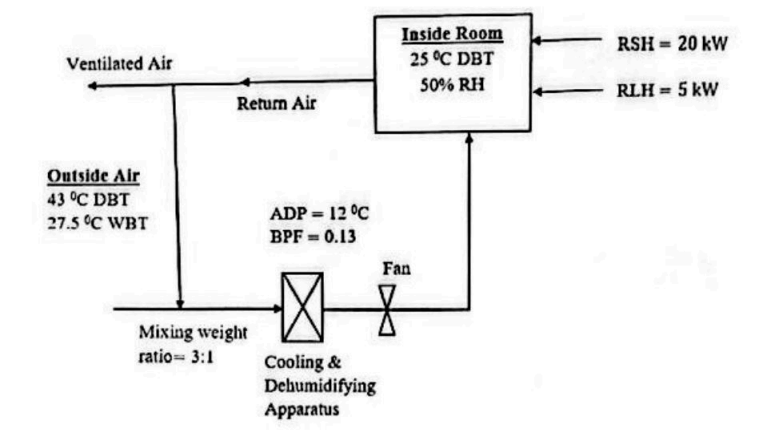
Remember: Opaque UA·ETD, glass $A(U\Delta T + SHGF \cdot SC)$, infiltration $m\,c\Delta T + m\,h_{fg}\Delta W$, people S+L, lights, appliances. Sum × 1.25.

B3 Summer AC with ventilation air and bypass factor (numerical) Must study

Very likely · 90% Hard

PAST-PAPER QUESTIONS

- **2024 Q5(c)** [15] A refrigeration plant is shown in figure. Determine: (i) Dehumidified air quantity, (ii) Sensible and latent heat gain due to outside air, (iii) Total refrigeration load of the plant.



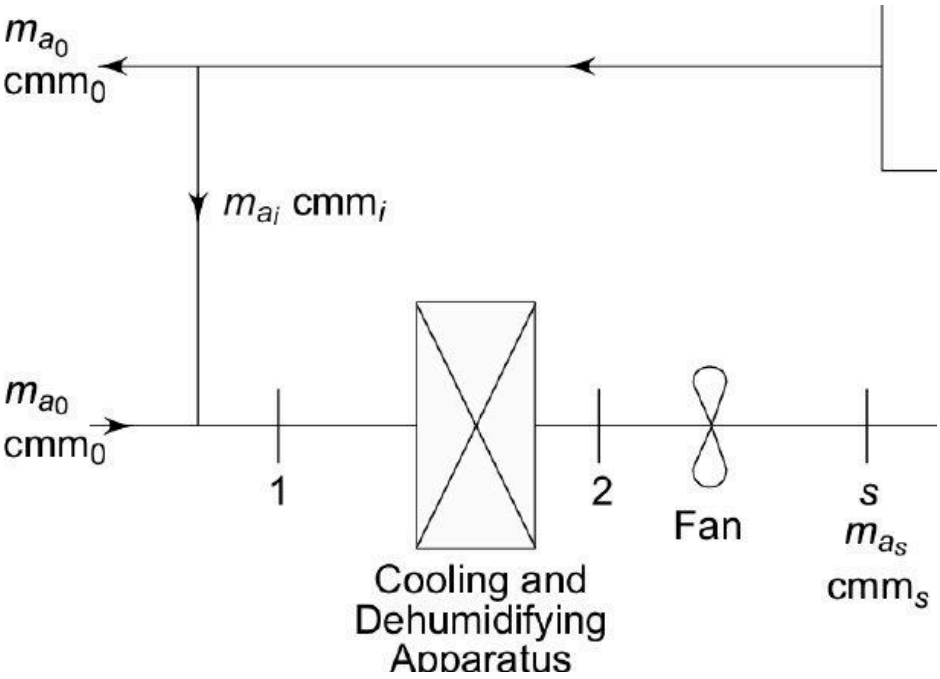
- **2023 Q5(d)** [11] Describe a summer air-conditioning with ventilation air-bypass factor.
- **2021 Q5(c)** [15] The room sensible heat gain and latent heat gain of a conditioned space are 40 kW and 15 kW respectively. The inside design conditions are 25 °C DBT (dry bulb temperature) and 55% RH (relative humidity). The return air is mixed with outside air before entering the cooling coil in the ratio of 3:1 by weight. Assuming the bypass factor of the cooling coil is 0.13, determine: (i) Apparatus dew point (ADP), (ii) Ventilation air in cmm, (iii) Total refrigeration load of the plant.

Before you start: the summer AC numerical in plain words

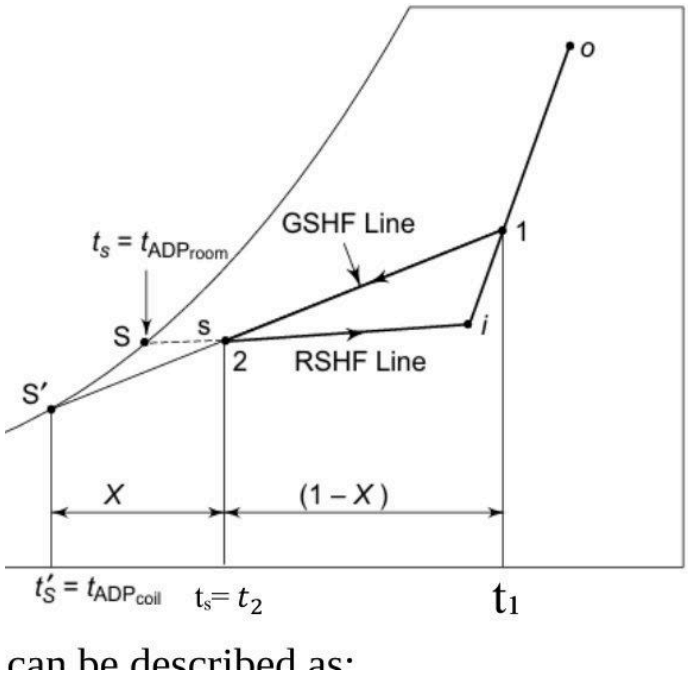
Room air (i) is mixed with some hot outside air (o) for freshness: the mix (1) is found by mass-weighted average. The cooling coil cools it towards its surface temperature, the **apparatus dew point (ADP)**, but a fraction X of air passes through untouched: the **bypass factor**. So coil exit t_2 = ADP (apparatus dew point) + X(t_1 − ADP).

Supply air must pick up the room sensible heat: m = RSH (room sensible heat) / [c_p(t_i − t_2)]. The coil must remove room load + outside air load.

Theory (2023 5d): summer AC with ventilation air and bypass factor



Return air i mixes with ventilation air o (state 1), is cooled in the coil (2) and supplied by the fan (class 06 slides)



can be described as:

RSHF (room sensible heat factor) line i-2 and GSHF (grand sensible heat factor) line 1-S′; supply state lies on the GSHF line at bypass distance X from ADP (class 06 slides)

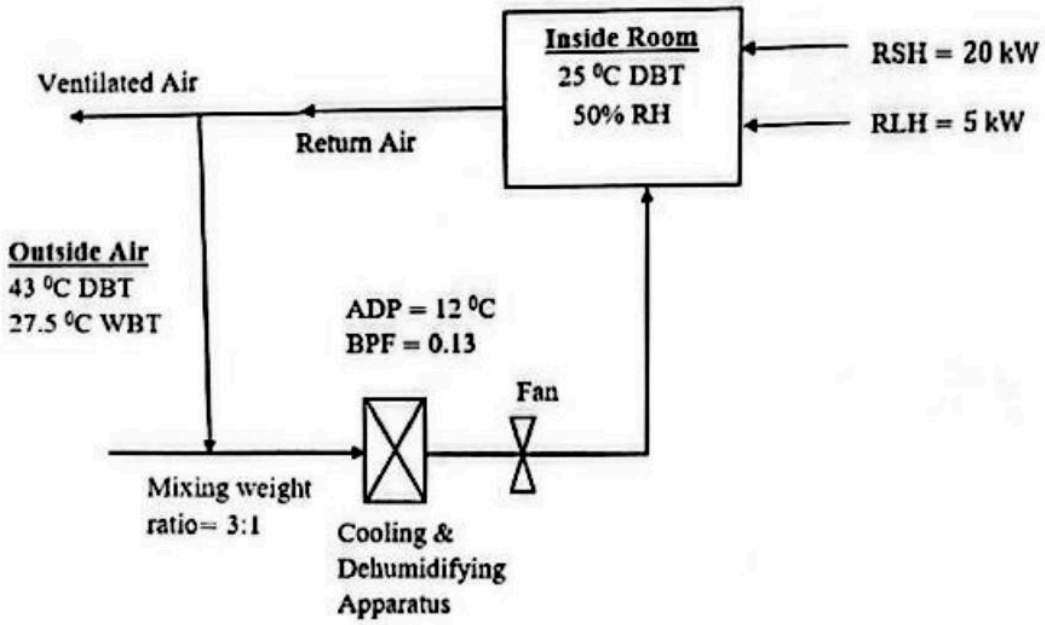
1. Outside air o (hot, humid) mixes with room return air i in a fixed ratio to give state 1 (mixing line, by mass ratio).
2. Mixed air passes over the coil with coil ADP t′$_s$. Because of bypass factor X, it leaves at 2, not at ADP: $t_2 = t_{ADP} + X(t_1 - t_{ADP})$. Line 1-S′ is the **GSHF line** (apparatus condition line).
3. Supply air 2 enters the room and picks up RSH and RLH (room latent heat) along the **RSHF line** 2-i.
4. Load on coil = room load + ventilation (outside air) load: **GTH (grand total heat) = RTH (room total heat) + OATH (outside air total heat)**.

$$(m_a)_d = RSH / [c_{pm}(t_i - t_{ADP})(1 - X)] = RSH / [c_{pm}(t_i - t_s)]$$

$$OASH \text{ (outside air sensible heat)} = m_o c_{pm}(t_o - t_i) \quad OALH \text{ (outside air latent heat)} = m_o h_{fg}(w_o - w_i) \quad OATH = m_o(h_o - h_i)$$

```
GTH = RSH + RLH + OATH = ma(h1 - h2);   GSHF = TSH (total sensible heat)/GTH
```

Numerical 1 (2024 5c): RSH 20 kW, RLH 5 kW, room 25 °C / 50%, OA 43 °C DBT / 27.5 °C WBT (wet bulb temperature), ADP 12 °C, X = 0.13, 3:1



Plant given in the 2024 question

Chart values (same as the class 06 example): outside w_o = 17 g/kg, h_o = 87.5 kJ/kg, v_o = 0.922 m³/kg; inside w_i = 10 g/kg, h_i = 50.8 kJ/kg; c_{pm} = 1.02, h_{fg} = 2500.

3:1 by weight → return 75%, outside 25%.

Mixed: t_1 = 0.75(25) + 0.25(43) = 29.5 °C; w_1 = 0.75(10) + 0.25(17) = 11.75 g/kg; h_1 = 0.75(50.8) + 0.25(87.5) = 60.0 kJ/kg

Coil exit: t_s = 12 + 0.13(29.5 − 12) = 14.3 °C

(i) Dehumidified air $(m_a)_d$ = 20 / [1.02(25 − 14.3)] = 1.83 kg/s (≈1.83 × 0.86 × 60 ≈ 94 m³/min)

Outside air m_o = 0.25 × 1.83 = 0.457 kg/s (= 0.457 × 0.922 = 0.42 m³/s)

(ii) OASH = 0.457 × 1.02 × (43 − 25) = 8.4 kW; OALH = 0.457 × 2500 × (0.017 − 0.010) = 8.0 kW

OATH = 0.457 × (87.5 − 50.8) = 16.8 kW

(iii) Total load GTH = RTH + OATH = (20 + 5) + 16.8 = 41.8 kW ≈ 11.9 TR (ton of refrigeration (3.5 kW))

Numerical 2 (2021 5c): RSH 40 kW, RLH 15 kW, room 25 °C / 55%, OA 35 °C / 65% RH (written on the paper), 3:1, X = 0.13

Chart: outside w_o = 23.4 g/kg, h_o = 95.2 kJ/kg, v_o = 0.905 m³/kg; inside w_i = 10.9 g/kg, h_i = 53.0 kJ/kg

Mixed: $t_1 = 0.75(25) + 0.25(35) = 27.5$ °C; $w_1 = 0.75(10.9) + 0.25(23.4) = 14.05$ g/kg

(i) ADP by trial: supply state 2 must satisfy both the bypass relation and the room ratio RSH/RLH = 40/15 = 2.67:

$t_2 = t_{ADP} + 0.13(27.5 - t_{ADP})$, $w_2 = w_{ADP} + 0.13(14.05 - w_{ADP})$, $1.0216(25 - t_2) / [2501(w_i - w_2)] = 2.67$

Try t_{ADP} = 11.5 °C (w_{ADP} = 8.45 g/kg): t_2 = 13.6 °C, w_2 = 9.18 g/kg → ratio = 1.0216(11.4)/(2501 × 0.00172) = 2.67 ✓

ADP ≈ 11.5 °C

$(m_a)_d = 40 / [1.0216(25 - 13.6)] = 3.42$ kg/s; $m_o = 0.25 \times 3.42 = 0.854$ kg/s

(ii) Ventilation air = 0.854 × 0.905 × 60 = 46.4 m³/min (cmm)

OATH = 0.854(95.2 − 53.0) = 36.1 kW

(iii) Total load = 40 + 15 + 36.1 = 91.1 kW ≈ 25.9 TR

Values were read from moist-air equations at 101.325 kPa; chart readings may differ by about 2-3%. If your copy of the 2021 paper has no outside condition, state "assumed 35 °C DBT, 65% RH" before solving.

Remember: Mix → coil (t_s = ADP + X(t_1 − ADP)) → room. m = RSH/[1.02(t_i − t_s)]. Total = RTH + $m_o(h_o - h_i)$.

Same answer, step by step with reasons

Step (write this)	Why / what it means
Mix 3:1 → t1 = 0.75 × 25 + 0.25 × 43 = 29.5 °C; h1 = 0.75 × 50.8 + 0.25 × 87.5 = 60.0 kJ/kg	Mixing = weighted average by mass (3 parts return, 1 part outside).
Coil exit t2 = ADP + X(t1 - ADP) = 12 + 0.13 × 17.5 = 14.3 °C	13 % of air bypasses the coil uncooled.
m_a = RSH / [$c_{pm}(t_i$ - t2)] = 20/(1.02 × 10.7) = 1.83 kg/s	Supply air warms from 14.3 to 25 °C absorbing the room sensible heat.
m_o = 0.25 × 1.83 = 0.457 kg/s	Outside-air share.
OATH = $m_o(h_o$ - $h_i)$ = 0.457 × 36.7 = 16.8 kW	Load from cooling the outside air down to room condition.
Coil load = RSH + RLH + OATH = 20 + 5 + 16.8 = **41.8 kW ≈ 11.9 TR**	Room load plus fresh-air load; divide by 3.5 for TR.

Memory trick: "Mix, Bypass, Mass, Load".

B10 Duct design rules, outlets, dampers, diffusers

Good to have · Likely · 60% · Easy

PAST-PAPER QUESTIONS

- **2023 Q8(b)** [7] What factors should be considered while selecting the location of supply air outlets?
- **2023 Q7(a)** [6] Write down the functions of the following: (i) Dampers; (ii) Diffusers.
- **2021 Q8(c)** [6] What are the general rules for duct design?
- **2021 Q8(b)** [8] What factors should be considered to locating the supply air outlets and return air inlets?
- **2020 Q6(b)** [8] What are the general rules of duct design?

Before you start: duct design rules

Keep air speed moderate (noise), use gradual bends and transitions (less pressure loss), dampers to balance flow, and diffusers/grilles that spread air without draughts.

Not in the uploaded slides; standard textbook points.

General rules for duct design (2021 8c, 2020 6b)

1. Run ducts as straight and short as possible; avoid sharp bends (use turning vanes or radius elbows, R/W ≥ 1.25).
2. Keep the aspect ratio of rectangular ducts low (≤ 4:1, ideally near 1:1 to 3:1).
3. Use velocities within limits to control noise: main ducts ~5-8 m/s in residences, up to 10-12 m/s in commercial buildings; branches lower.
4. Change cross-section gradually (taper angle ≤ 15-20°) to reduce dynamic losses.
5. Seal joints to limit leakage; insulate ducts through unconditioned spaces.
6. Provide dampers for balancing and fire dampers where required; allow access for cleaning.
7. Balance the system so each outlet gets its design airflow; keep the longest run's pressure loss reasonable.

Factors for locating supply outlets and return inlets (2023 8b, 2021 8b)

- Throw, spread and drop of the air jet so air reaches the occupied zone at 0.15-0.25 m/s without draught.
- Location of heat sources and loads: put outlets near windows and outer walls to offset gains/losses.
- Ceiling height and room shape; furniture and obstructions.
- Avoid short-circuiting: returns placed away from supplies so air sweeps the room.
- Noise level and appearance; mixing and temperature uniformity; cooling vs heating mode (cool air from high level, warm from low level).
- Returns: near the floor or at high level for hot air, away from people, sized for low velocity (< 2-4 m/s).

Functions of dampers and diffusers (2023 7a)

- **Dampers:** control or shut off airflow in ducts: volume control and system balancing, mixing outdoor and return air, fire and smoke isolation (fire dampers), backdraft prevention.
- **Diffusers:** supply outlets that spread air into the room in a pattern, mix it quickly with room air, and lower its velocity so that there is no draught or noise.

Remember: Duct: short, straight, low aspect ratio, sealed, balanced. Outlets: throw, spread, no short-circuit.

B11 Sol-air temperature, window solar gain, infiltration

Good to have

Possible · 50% Medium

PAST-PAPER QUESTIONS

- **2023 Q8(a)** [4] What is meant by sol-air temperature?
- **2021 Q7(d)** [8] What is sol-air temperature? How to estimate the solar heat gain through building window?
- **2021 Q8(a)** [7] What is infiltration? How to estimate infiltration rate using Crack method?

Before you start: solar gains

A sunlit wall acts as if the outside air were hotter: **sol-air temperature** = $T_{air} + \alpha q/h_o$. Windows let sunshine straight in. Infiltration (air leaking through cracks) adds both sensible and latent load.

Sol-air temperature (2023 8a, 2021 7d)

The **sol-air temperature** is the fictitious outdoor air temperature that, with no solar radiation, would give the same heat flow into an outside surface as the actual combination of outdoor air temperature, incident solar radiation and long-wave sky exchange.

$t_{sa} = t_o + \alpha\, I_t/h_o - \varepsilon\Delta R/h_o$ (α = absorptivity, I_t = total solar radiation, h_o = outside film coefficient; last term ≈ 4 K for roofs, 0 for walls)

Heat gain through a wall or roof then is $Q = U\,A\,(t_{sa} - t_i)$ (steady), and CLTD (cooling load temperature difference)/ETD values are derived from it.

Solar heat gain through a window (2021 7d)

$Q_{glass} = A[U_{glass}(T_o - T_i) + \text{SHGF (solar heat gain factor)}_{max} \times \text{SC (shading coefficient)} \times \text{CLF (cooling load factor)}]$

- First term: conduction through the glass.
- SHGF: solar heat gain factor from tables (orientation, month, hour, latitude).
- SC: shading coefficient = gain through this glass / gain through standard clear glass; depends on glass type and blinds.
- CLF: cooling load factor for the time lag of radiant heat stored in room surfaces.

Infiltration and crack method (2021 8a)

Infiltration is the uncontrolled flow of outdoor air into a building through cracks around windows and doors and other unintended openings (air leakage), caused by wind and stack pressure.

Crack method: infiltration rate = (crack length) × (leakage per metre of crack) for the windward windows and doors.

$\dot{V} = \Sigma\,(L_{crack} \times q_{per\ m})$ m³/h, where q depends on window type and the wind speed (pressure difference), from tables

Then $Q_s = m\, c_{pm}(T_o - T_i)$ and $Q_L = m\, h_{fg}(W_o - W_i)$, with $m = \rho \dot{V}$. The simpler **air change method** uses $\dot{V}$ = ACH (air changes per hour) × room volume / 3600 m^3/s (used in the 2024 cooling load problem).

Remember: Sol-air = air temp + absorbed sun ÷ h_o. Crack method = crack length × leakage per metre.